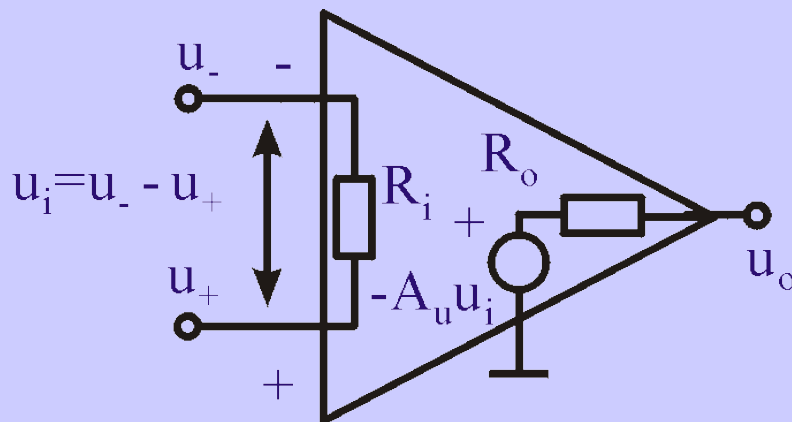
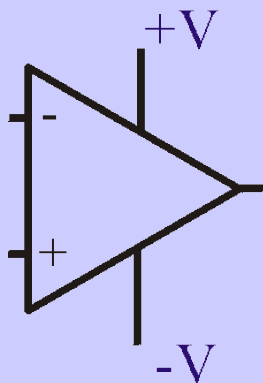


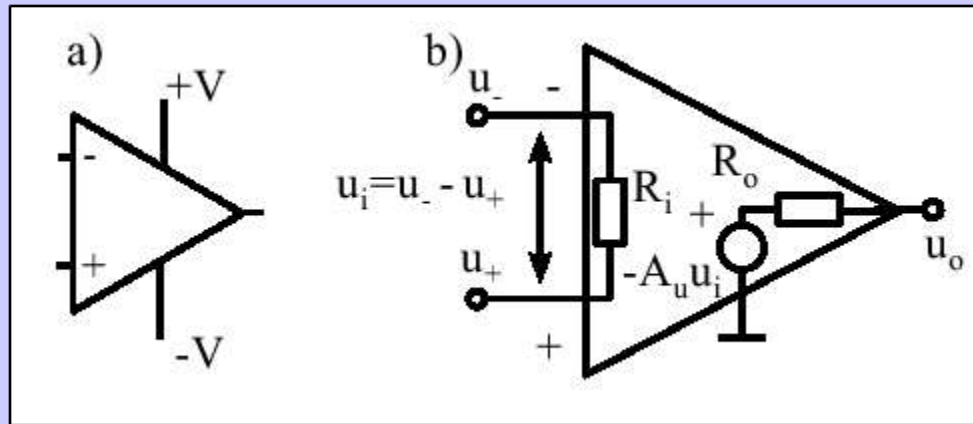
Operationsförstärkare

Operationsförstärkaren är elektronikens arbetsmyra: de finns i de flesta apparater

symbol på kretsschema



1. Operationsförstärkarens utgång försöker hålla dess skillnaden mellan ingångarna (+ och -) lika med noll med hjälp av negativ återkoppling.
2. Ingen ström i operationsförstärkarens ingångar (+ och -) dvs ingångsimpedansen är oändlig.



Ideaalisen operaatiovahvistimen (vrt. kuva 5.1b)

- sisäänmenoiresistanssi² $R_i \rightarrow \infty$,
- ulostuloresistanssi $R_o = 0$,
- jännitevahvistus $A_u \rightarrow \infty$,
- ulostulojännite u_o on äärellinen, mistä seuraa ehto $u_i = 0$,
- operaatiovahvistimen kaistanleveys on ääretön,
- lopuksi $u_+ = u_- \Rightarrow u_o = 0$ riippumatta jännitteestä u_+ .

Grundkopplingar

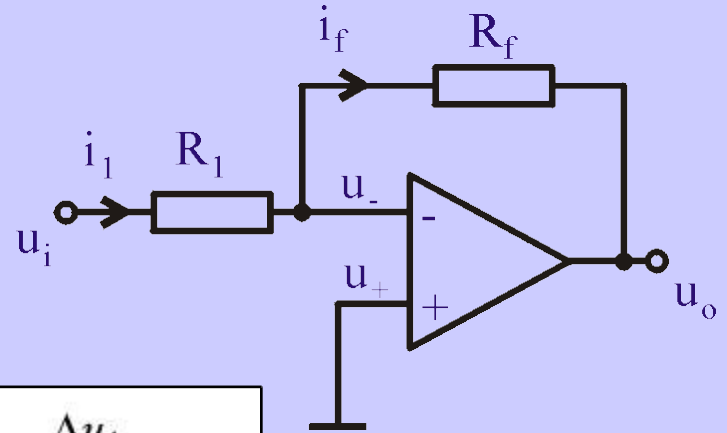
Inverterade förstärkare

$$i_1 = \frac{u_i - u_-}{R_1} = \frac{u_i}{R_1} ,$$

$$u_o = u_- - R_f \cdot i_f = 0 \text{ V} - R_f \cdot i_1 = -R_f \frac{u_i}{R_1} .$$

$$A_V = \frac{u_o}{u_i} = -\frac{R_f}{R_1} .$$

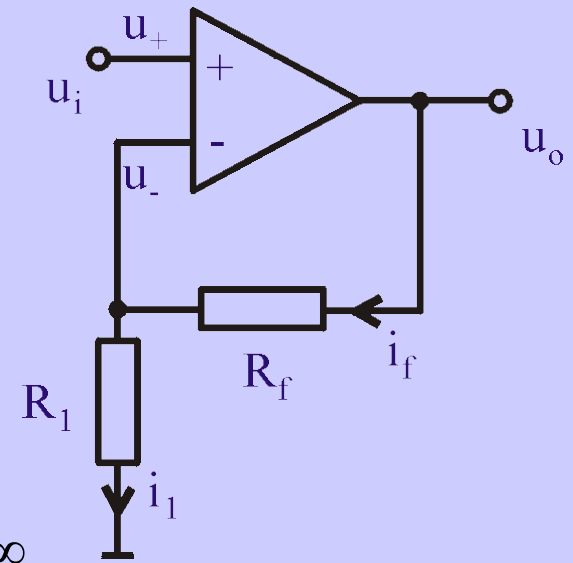
$$Z_i = \frac{\Delta u_i}{\Delta i_1} = R_1 .$$



Icke-inverterande förstärkare

$$u_i = u_- = \frac{R_1}{R_1 + R_f} \cdot u_o .$$

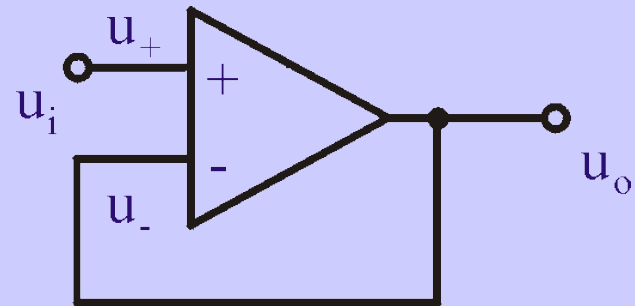
$$A_V = \frac{u_o}{u_i} = \frac{R_1 + R_f}{R_1} = 1 + \frac{R_f}{R_1} .$$



För en ideal förstärkare $Z_i \rightarrow \infty$

Spänningsföljare (buffert)

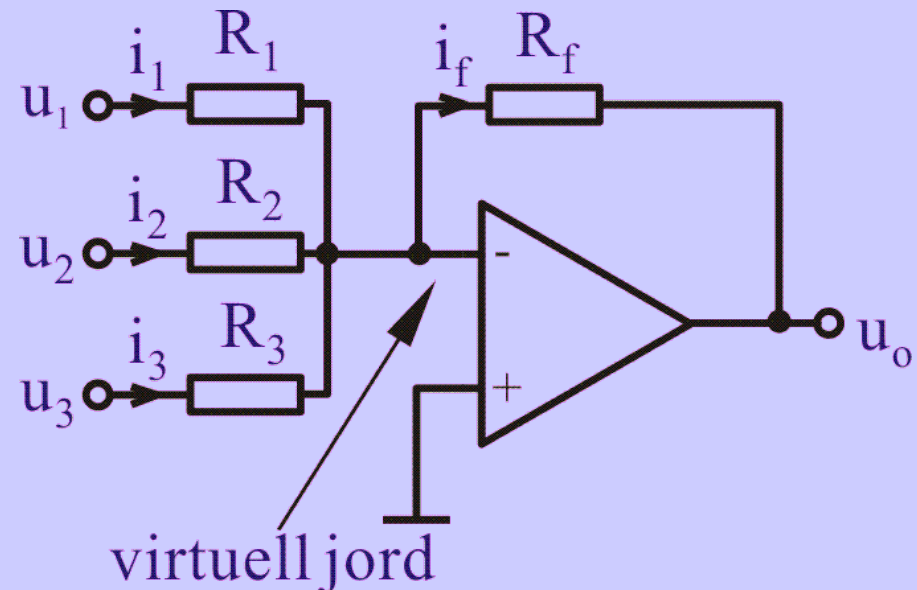
- hög ingångsimpedans
- specialfall av den föregående!



Summerare

- adderar signaler

$$u_o = R_f \cdot i_f = \frac{R_f}{R_1} \cdot u_1 + \frac{R_f}{R_2} \cdot u_2 + \frac{R_f}{R_3} \cdot u_3$$



Differentialförstärkare

- utgången är en förstärkning av ingångarnas spänningsskillnad

$$u_- = u_+ = \frac{R_4}{R_4 + R_2} \cdot u_2 \cdot$$

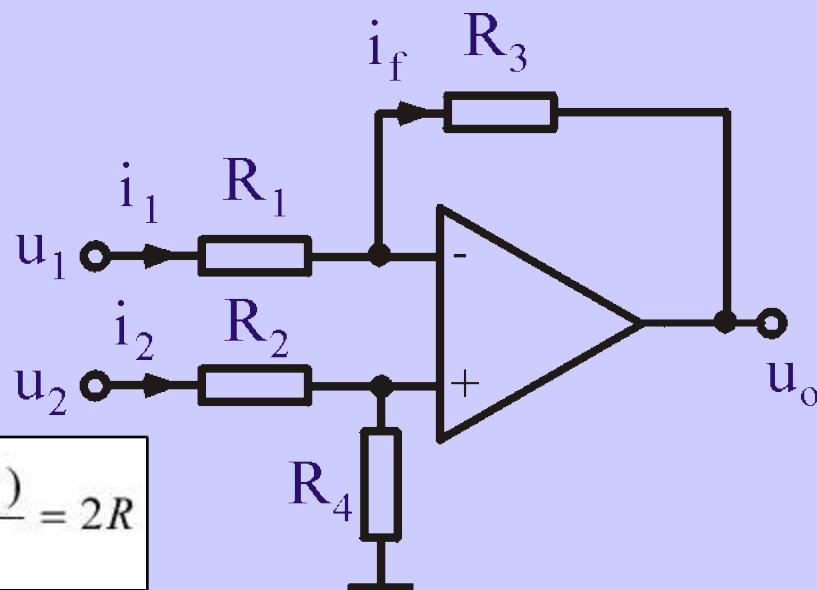
$$i_f = i_1 = \frac{u_1 - u_-}{R_1} = \frac{u_1 - \frac{R_4}{R_4 + R_2} \cdot u_2}{R_1} = \frac{u_1}{R_1} - \frac{R_4}{R_1(R_4 + R_2)} \cdot u_2 \cdot$$

$$\begin{aligned} u_o &= u_- - R_3 \cdot i_f = u_+ - R_3 \cdot i_f \\ &= \frac{R_4}{R_4 + R_2} \cdot u_2 + \frac{R_3 R_4}{R_1(R_4 + R_2)} \cdot u_2 - \frac{R_3}{R_1} \cdot u_1 \\ &= \frac{R_1 R_4 + R_4 R_3}{R_1(R_4 + R_2)} \cdot u_2 - \frac{R_3}{R_1} \cdot u_1 \cdot \end{aligned}$$

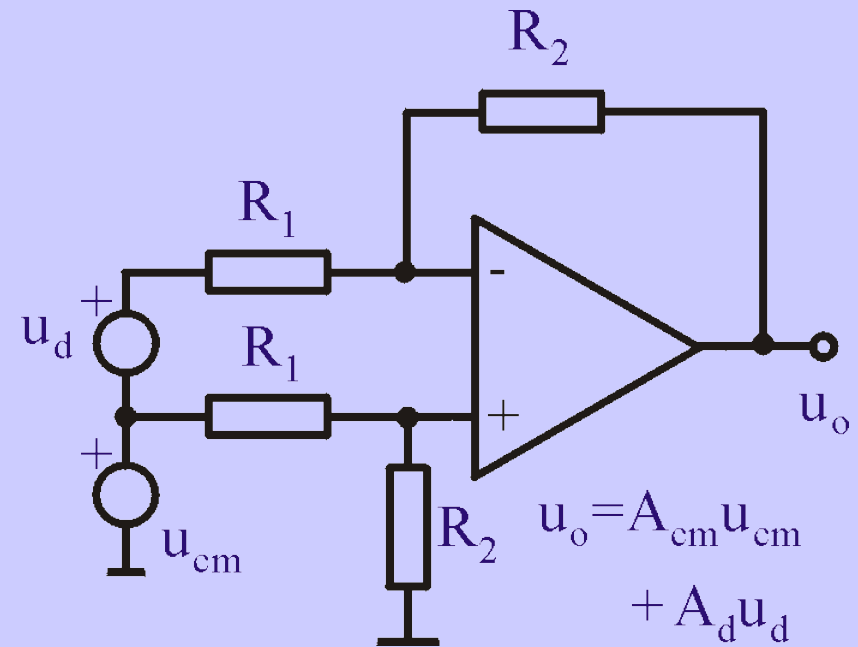
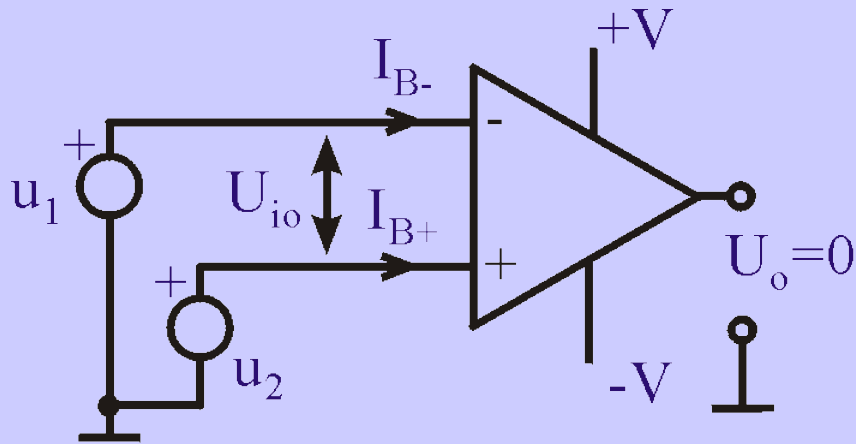
ofta $R_1 = R_2 = R$ ja $R_3 = R_4 = R_f$ varvid

$$u_o = \frac{R_f}{R} (u_2 - u_1) \cdot$$

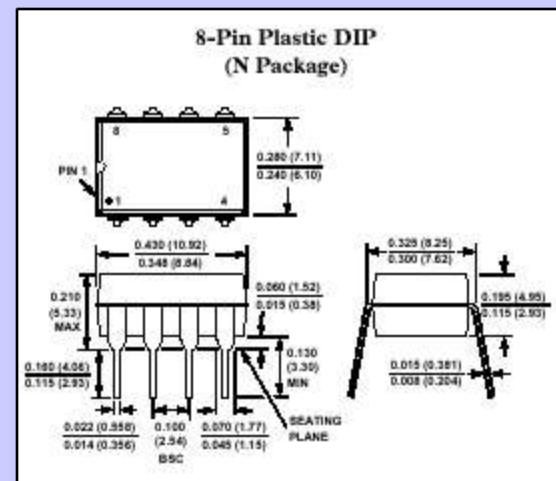
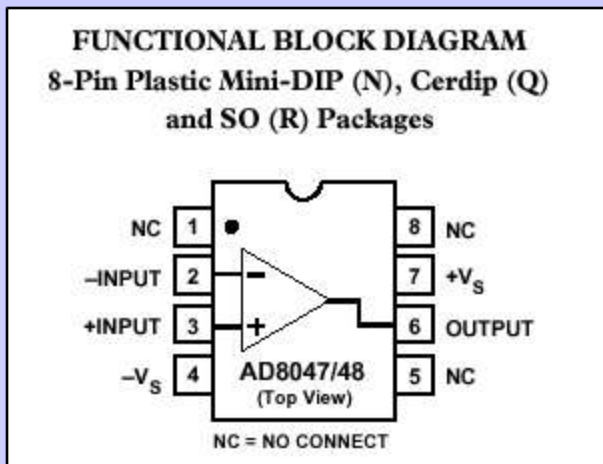
$$Z_i = \frac{\Delta(u_2 - u_1)}{\Delta i} = 2R$$

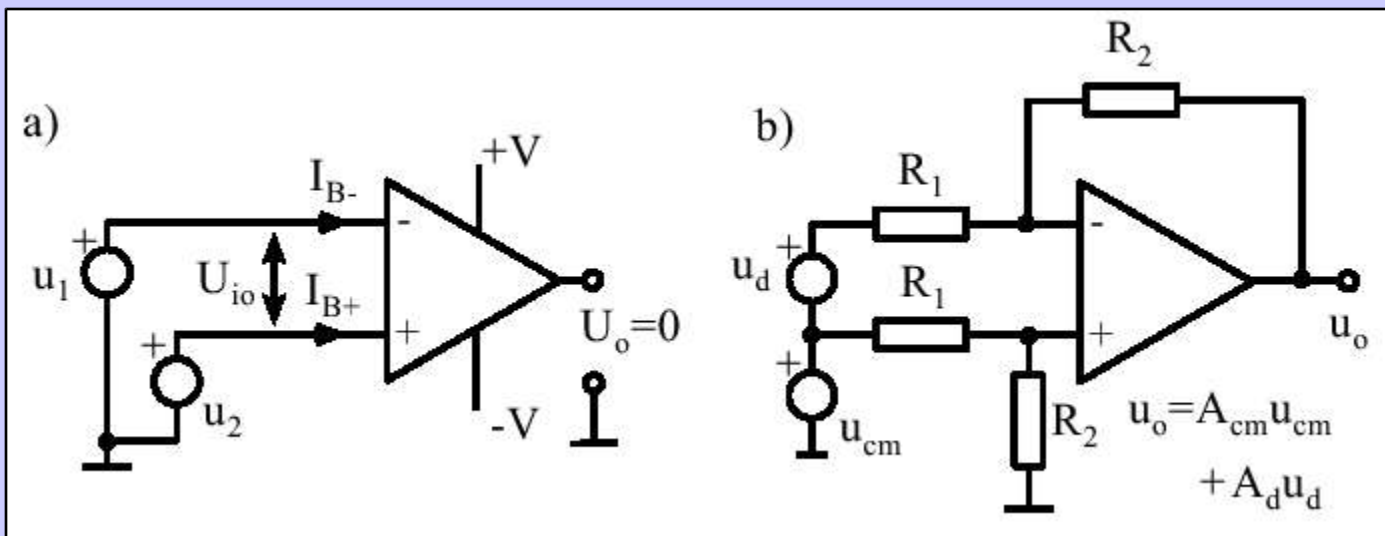


Operationsförstärkaren i praktiken



- i praktiken har operationsförstärkaren många parametrar, som avviker från de ideala



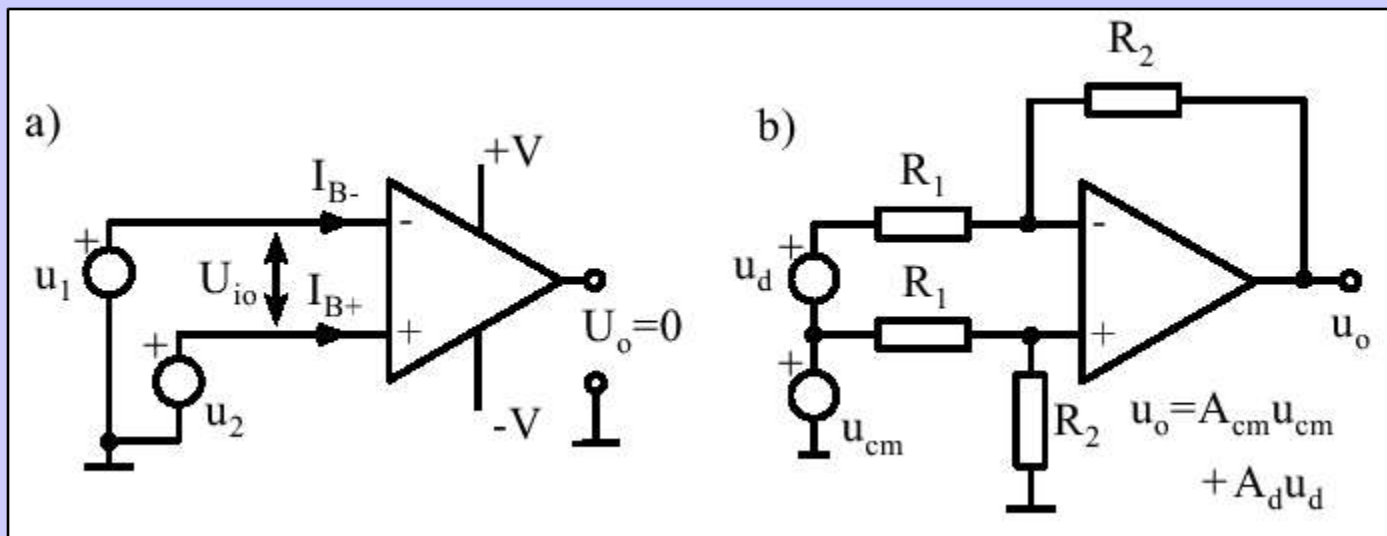


1) *Input bias current.* Sisäänmenon biasvirta I_B on puolet kuvaan merkittyjen virtojen summasta eli $I_B \equiv \frac{1}{2}(I_{B-} + I_{B+})$, kun $U_o = 0$.

2) *Input offset current.* Sisäänmenojen biasvirtaero I_{io} on kuvaan merkittyjen virtojen erotus $I_{io} = I_{B+} - I_{B-}$, kun $U_o = 0$.

3) *Input offset current drift.* Sisäänmenojen biasvirtaeron siirtymä $\frac{\Delta I_{io}}{\Delta T}$ kuvaa sisäänmenojen biasvirtaeron lämpötilariippuvuutta.

4) *Input offset voltage.* Sisäänmenojen jännite-ero U_{io} on napojen välille asetettava jännite, joka saa aikaan vahvistimen balansoitumisen ($U_o \rightarrow 0$ V).



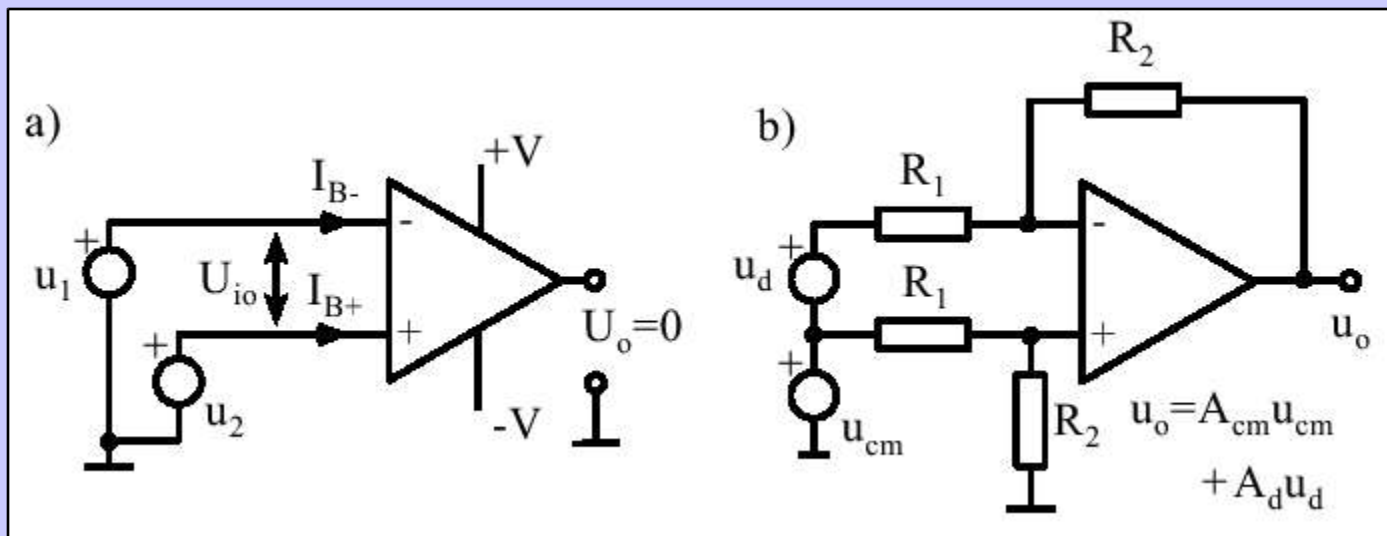
5) *Input offset voltage drift.* Sisäänmenojen jännite-eron siirtymä $\frac{\Delta U_{io}}{\Delta T}$ kuvaa sisäänmenojen jännite-eron lämpötilariippuvuutta.

6) *Output offset voltage.* Ulostulojännitteen jännite-ero U_{∞} on ulostulojännitteen arvo, kun vahvistimen sisäänmenot on maadoitettu.

7) *Input common-mode range.* Sisäänmenon yhteismuotoalue määrää vahvistimien lineaarisuusalueen sisäänmenojen suhteen.

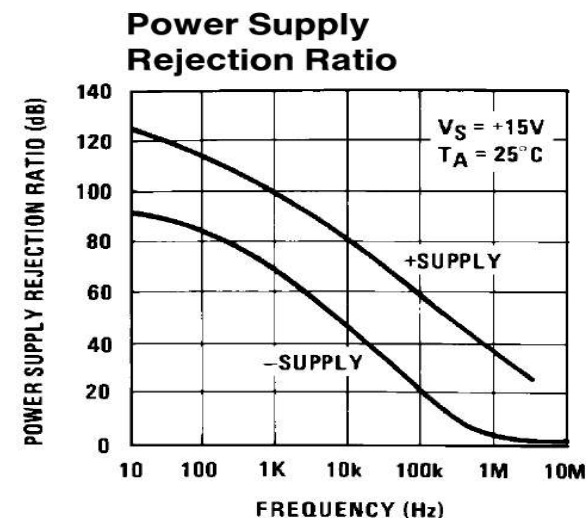
8) *Input differential range.* Sisäänmenon erosignaali-alue kertoo suurimman jännitteen, joka voidaan asettaa vahvistimen sisäänmenojen välille.

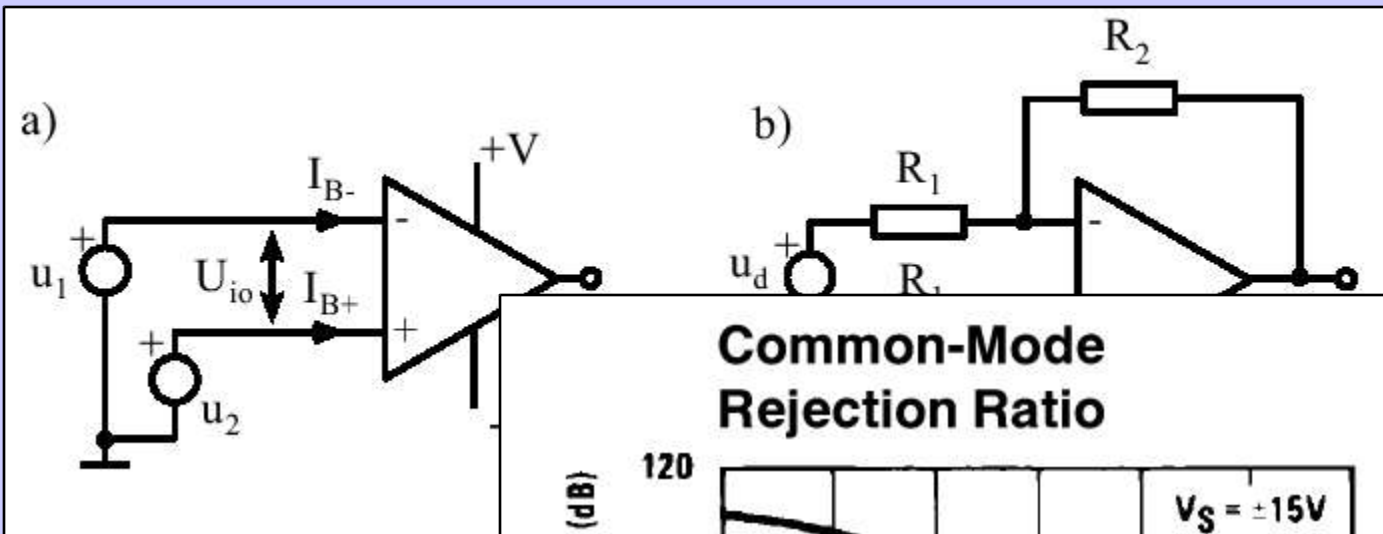
9) *Output voltage range.* Ulostulon jännitealue antaa suurimman säröttömän jännitteen heilahtelualueen kullakin kuormalla.



10) *Full-power bandwidth*. Täyden tehon kaistanleveys kertoo suurimman taajuuden, jolla saadaan aikaan kohdan 9 jänniteheilahtelu.

11) *Power-supply rejection ratio (PSRR)*. Teholähtö ilmoitetaan sisäänmenosignaalin muutoksen suhteena. Se kuvaa vahvistimen ulostulon riippumattomuutta. PSRR pienenee taajuuden kasvaessa.





12) *Common-mode rejection* suhde kuvaa vahvistimen kyseen poikkeamaan.

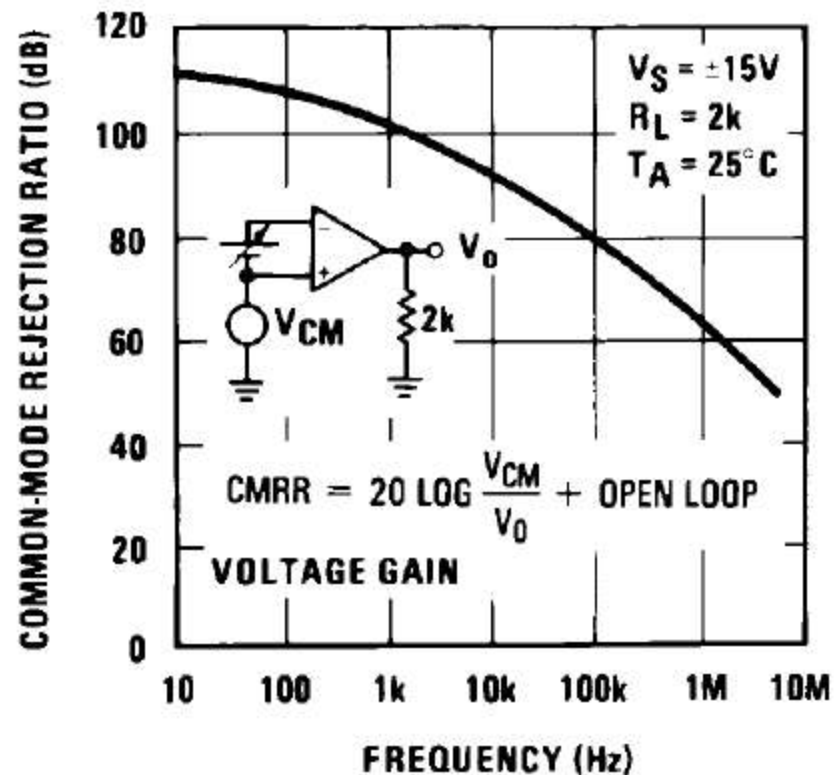
$$CMRR = \left| \frac{A_d}{A_{cm}} \right|$$

CMRR

Käytännön operaatiovahvistin pienenee taajuuden kasvaessa instrumentointivahvistimia (

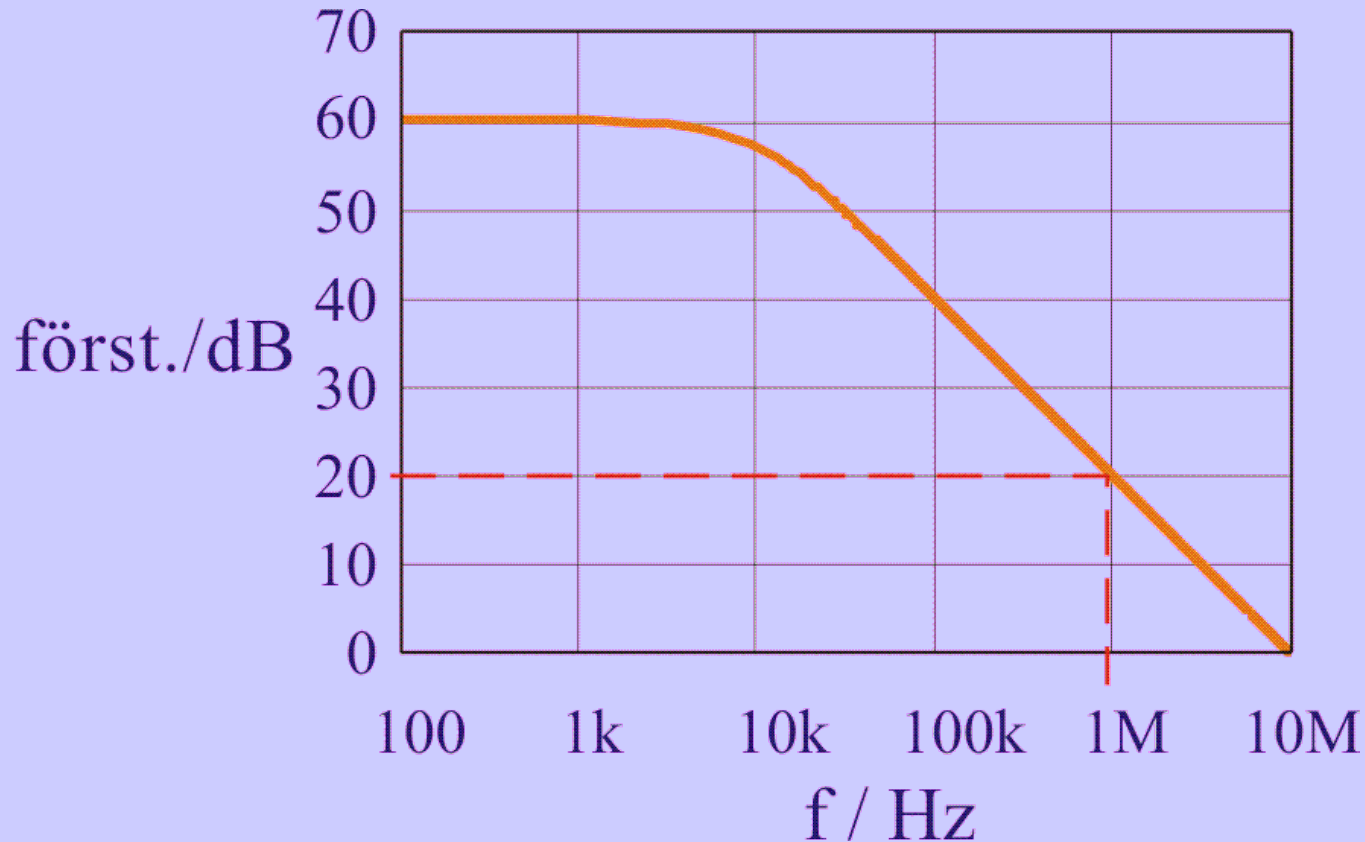
13) *Slew rate*. Muutosnopeuden (V/s)suurel

Common-Mode Rejection Ratio

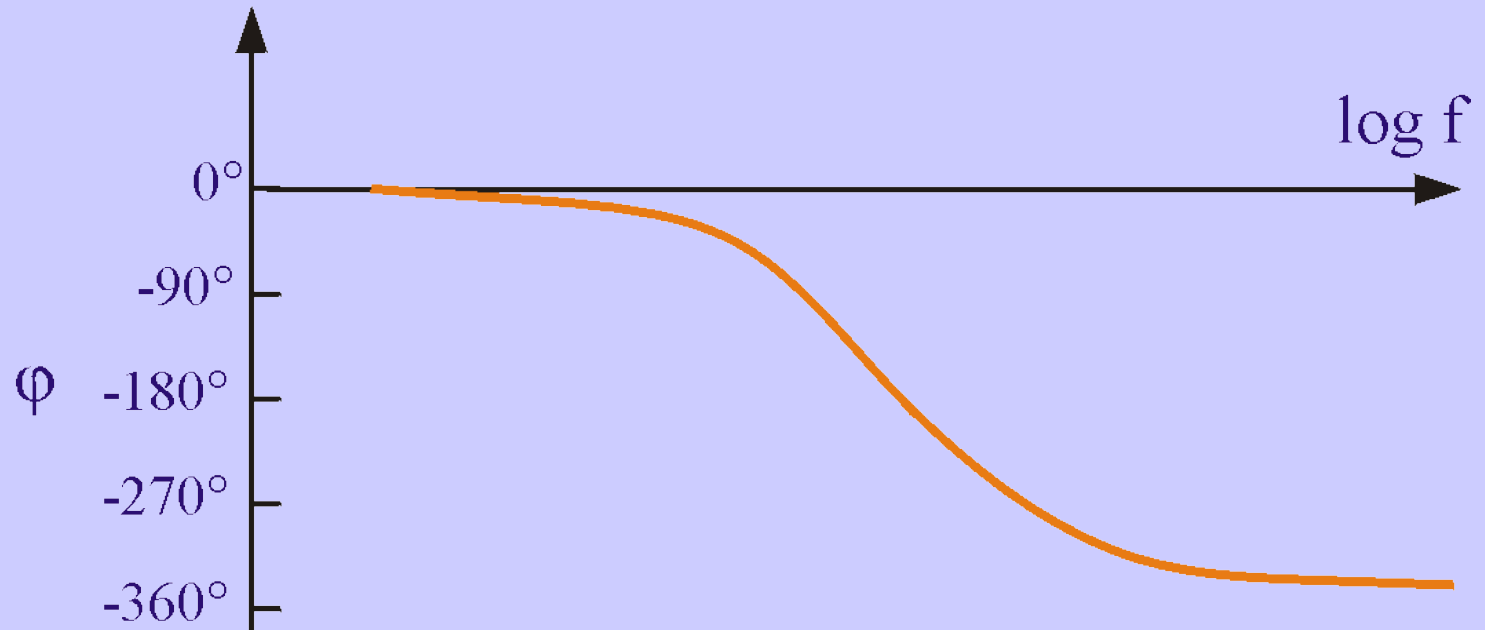


Bandbredd

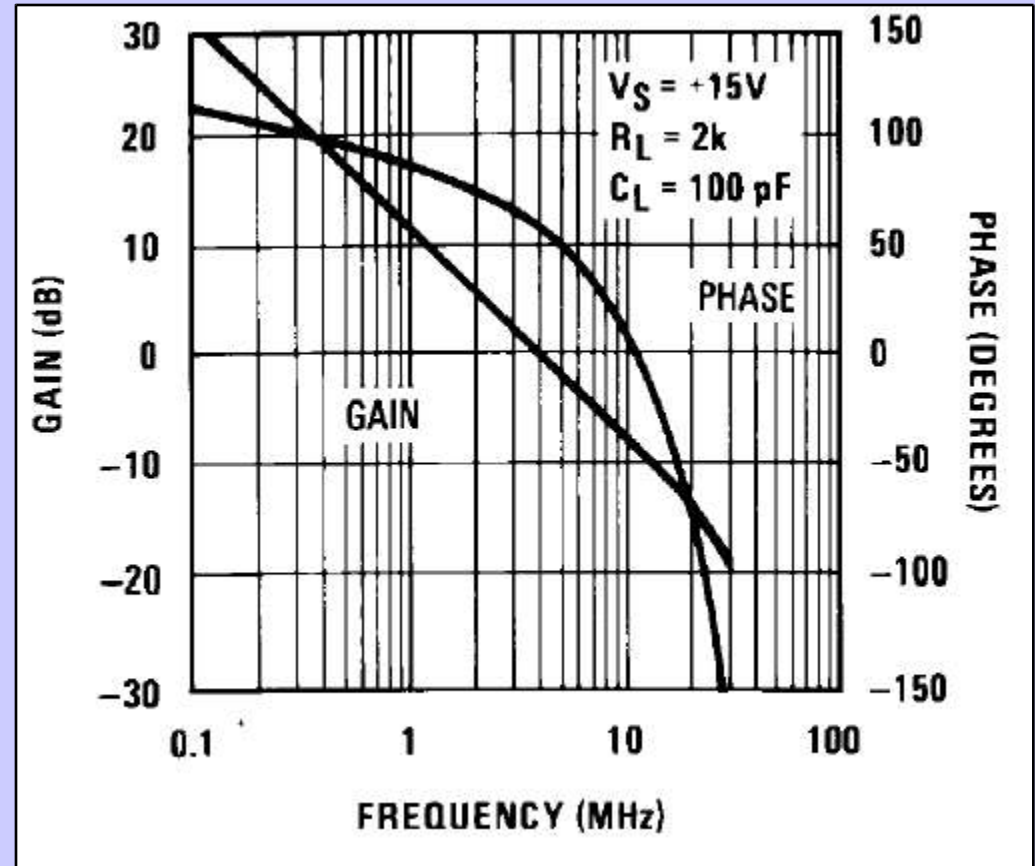
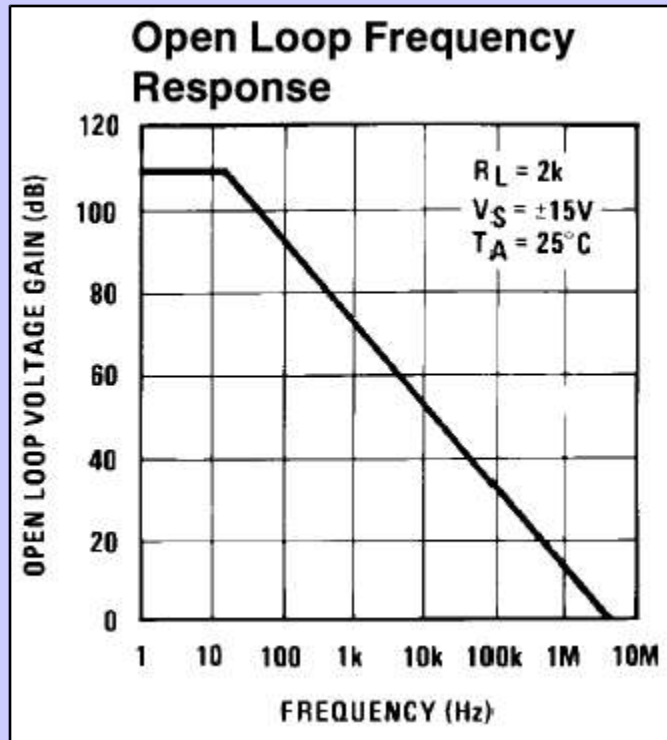
- en reell operationsförstärkare har ändlig bandbredd
- förstärkarkretsens frekvensbandbredd kan regleras med hjälp av återkoppling



Utgångens fasskift

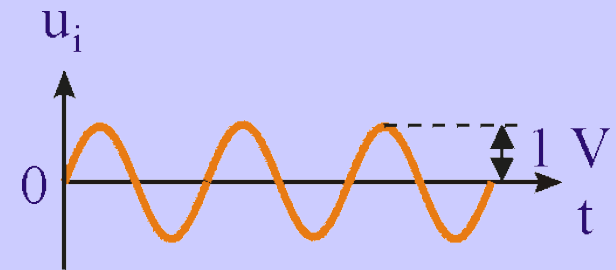
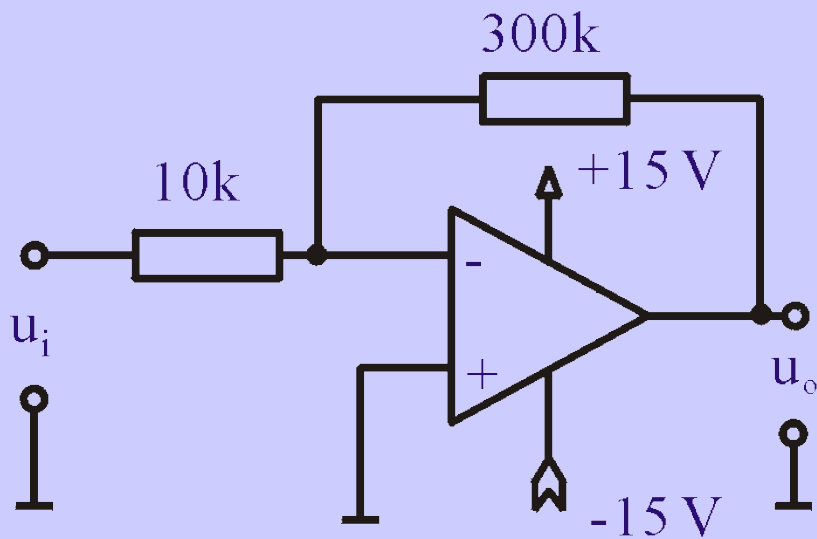


Förstärkning och utgångens fasskift (operationsförstärkaren LF351)

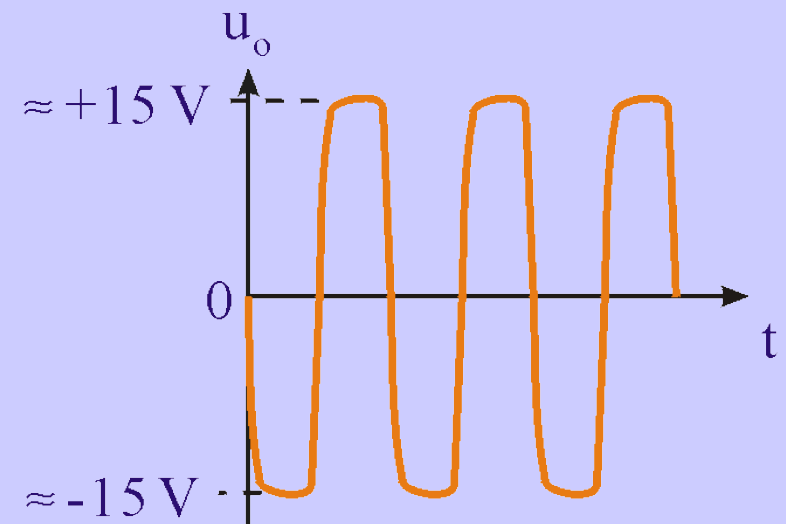


Bruksspänningar

Inverterande förstärkare ($A_v=30$)

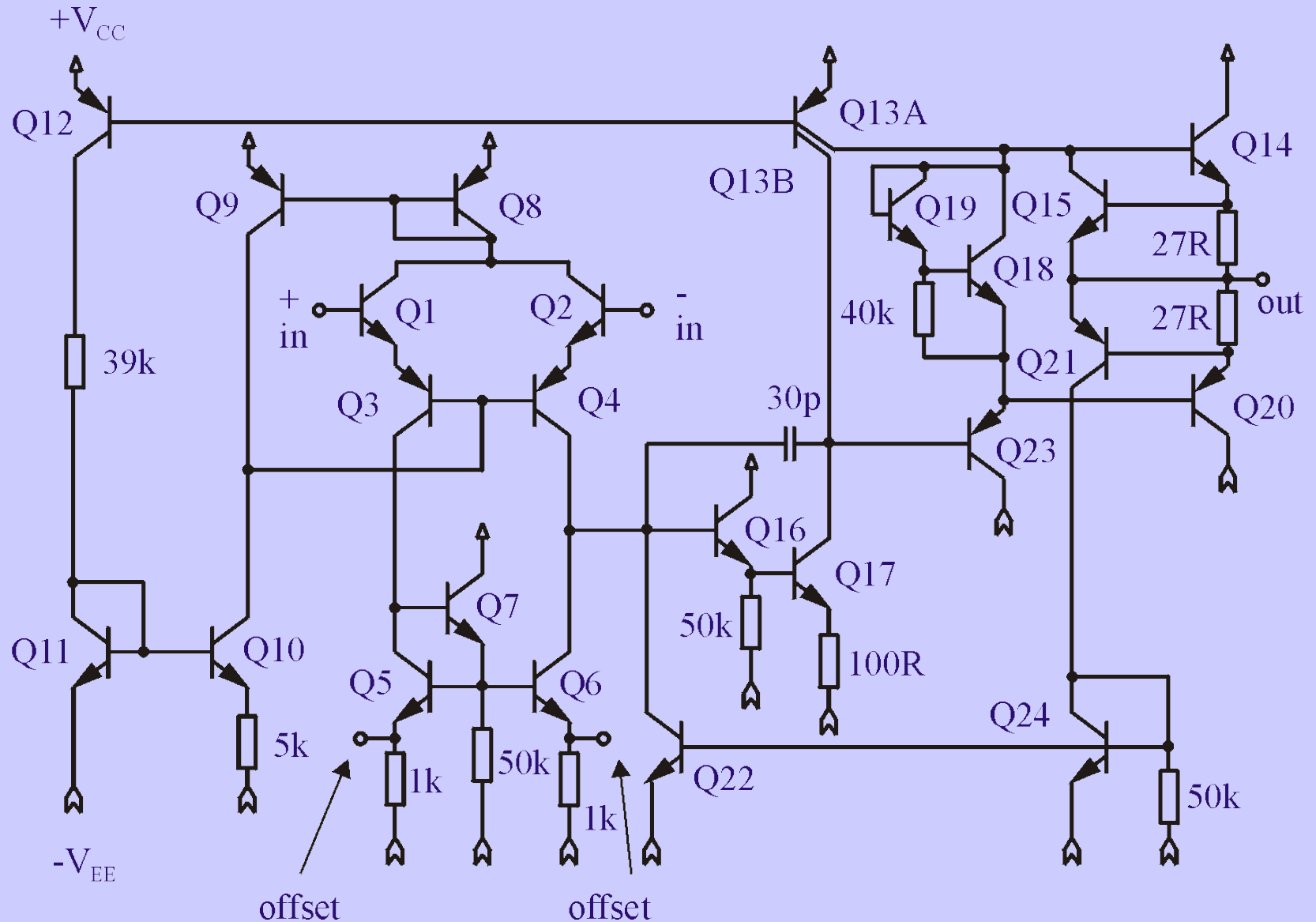


ingångssignal



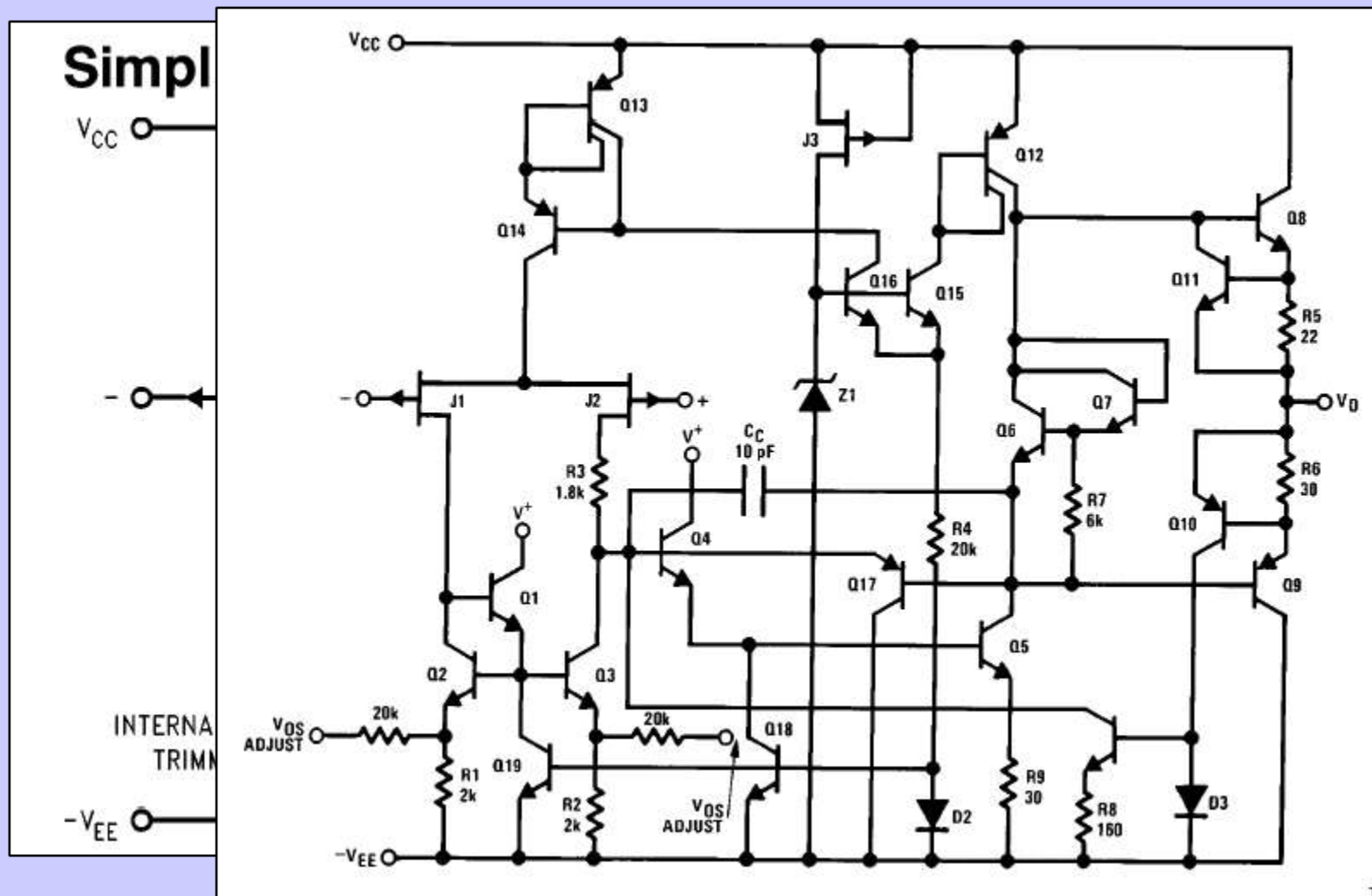
utgångssignal

Operationsförstärkarens uppbyggnad (741)

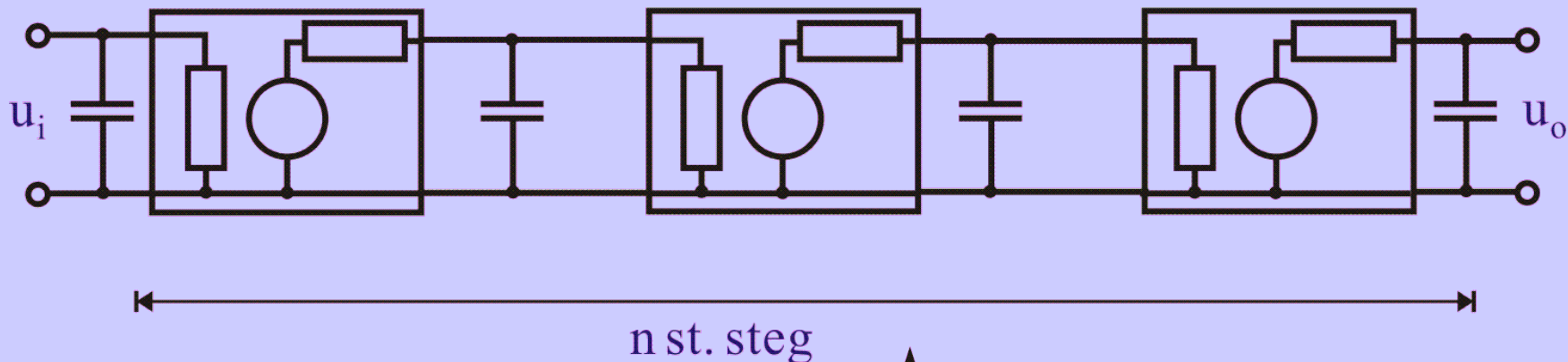


Operaatiovahvistimia voidaan valmistaa myös kanavatransistoreista. Yleisin tapa on rakentaa sisäänmenon differentiaalivahvistin JFET:ien (tai MOSFET:ien) avulla, jolloin operaatiovahvistimen sisäänmenoimpedanssi kasvaa moninkertaiseksi bipolaariseen toteutukseen verrattuna. Bipolaaristen operaatiovahvistimien sisäänmenoimpedanssit vaihtelevat muutamasta sadasta kilo-ohmista muutamaan megaohmiin, kun taas JFET -operaatiovahvistimen sisäänmenoimpedanssi on yleensä luokkaa 10^{12} ohmia (voi olla joskus jopa 10^{16} ohmia). Kanava- ja bipolaaritransistoreita sisältävien vahvistimien yhteydessä käytetään nimityksiä BIFET ja BIMOS.

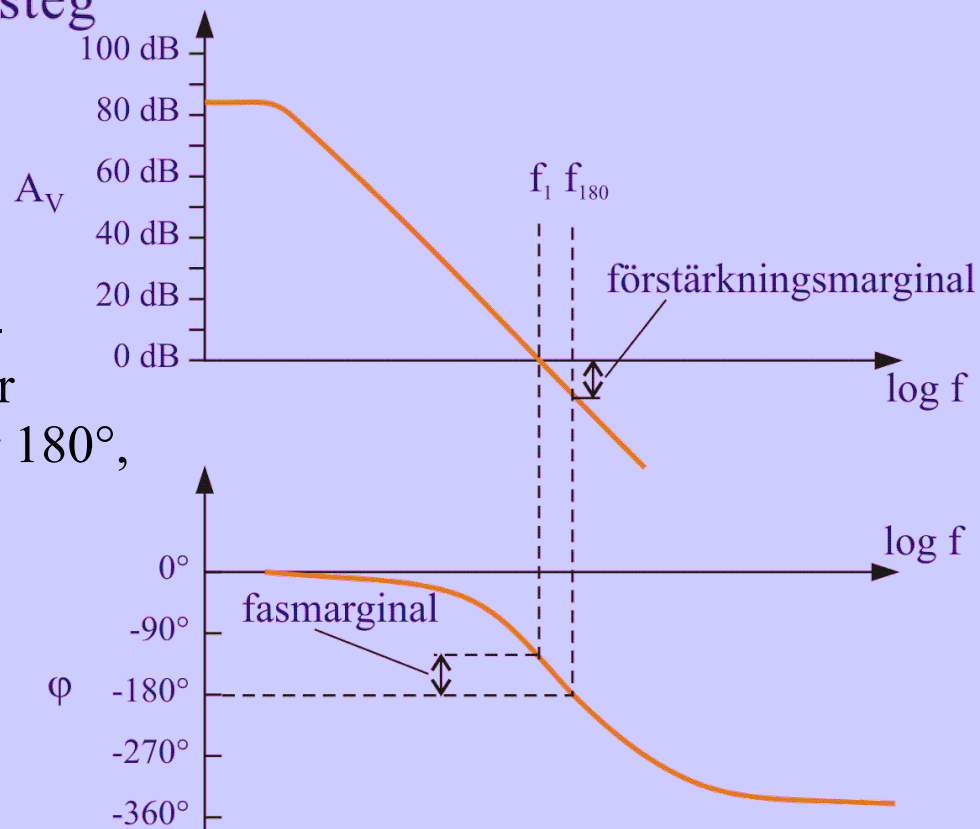
LF351 Wide Bandwidth JFET Input Operational Amplifier



Kompensering

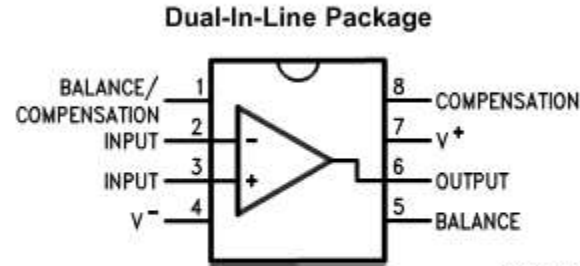


- en operationsförstärkare som är återkopplad bör ha en förstärkning som är mindre än ett då utgångens fasskift är 180° , annars börjar den att oskillera



LM101

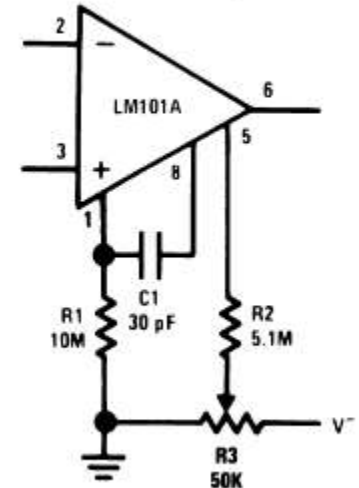
Connection Diagrams (Top View)



DS007752-4

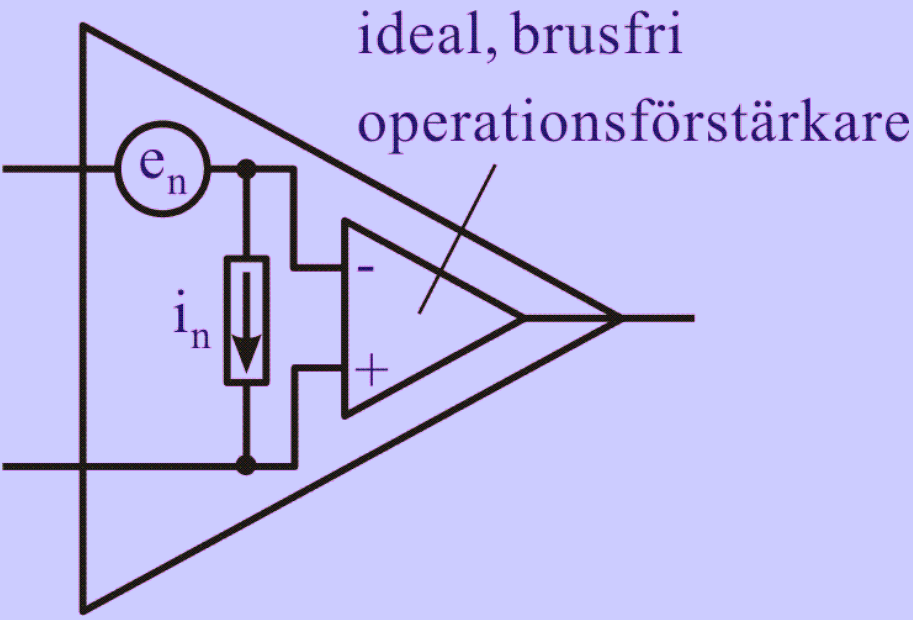
Order Number LM101AJ, LM101J/883 (Note 10),
LM201AN or LM301AN
See NS Package Number J08A or N08E

Standard Compensation and Offset Balancing Circuit



and output, no latch-up when the common mode range is exceeded, and freedom from oscillations and compensation with a single 30 pF capacitor. It has advantages over internally compensated amplifiers in that the frequency compensation can be tailored to the particular application. For example, in low frequency circuits it can be over-compensated for increased stability margin. Or the compensation can be optimized to give more than a factor of ten improvement in high frequency performance for most applications.





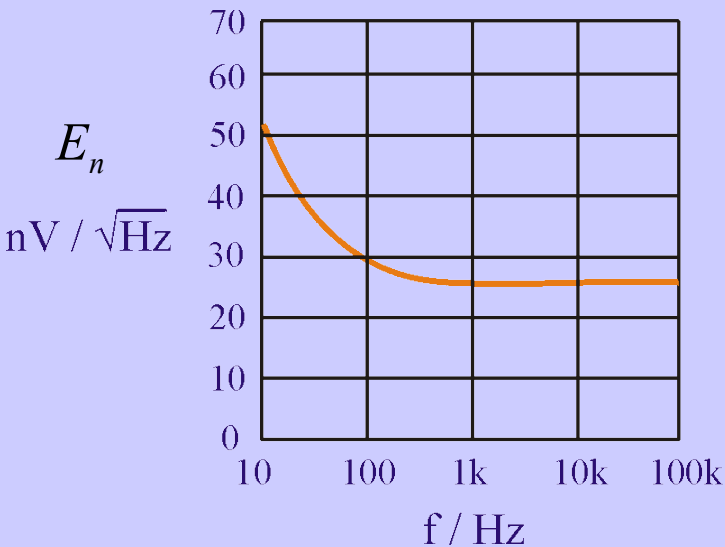
signalbrusförhållandet

$$S / N = \frac{P_s}{P_n}$$

signalens effekt

brusets effekt

$$E_n = \sqrt{\int_{f_1}^{f_2} e_n^2 df} = e_n \sqrt{f_2 - f_1}$$



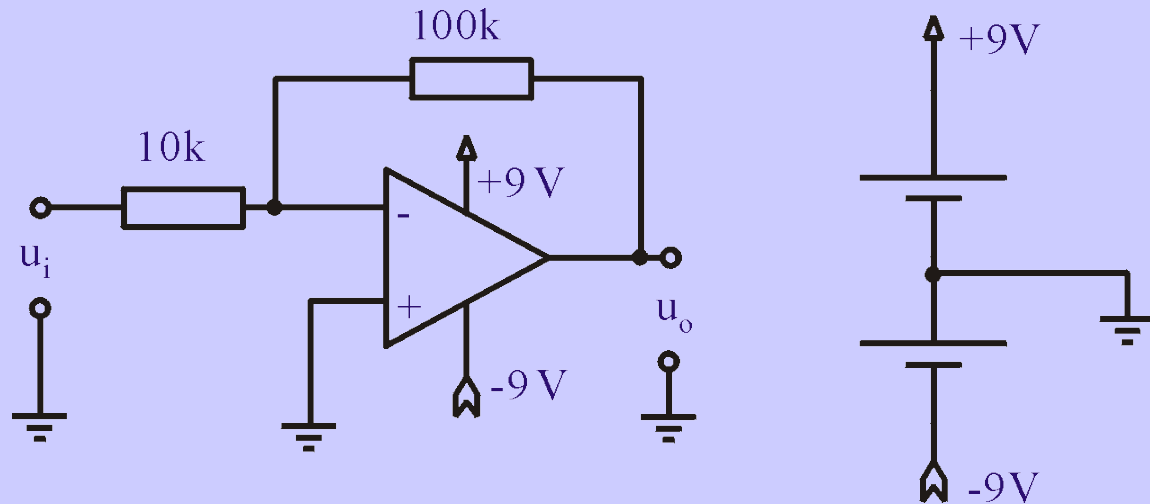
brusspänningens beror
av frekvensen

Mera info t.ex.

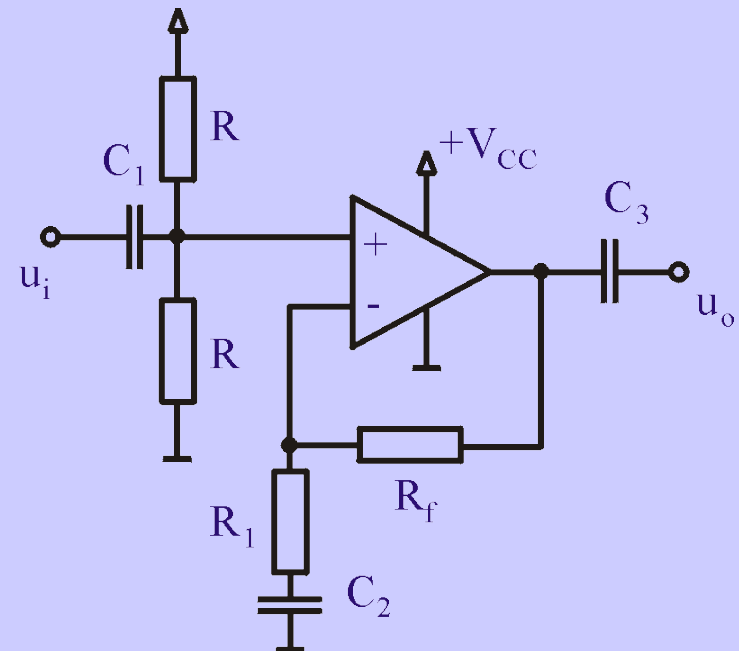
Analog Dialogue

Förstärkarkopplingar

Tvåsidig matarspänning
med hjälp av batterier



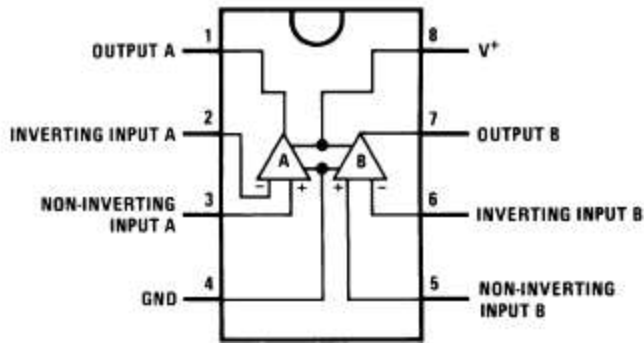
Ei-käntävän vahvistimen kytKentä
yksipuolisella käyttöjännitteellä



LM158/LM258/LM358/LM2904

Low Power Dual Operational Amplifiers

DIP/SO Package

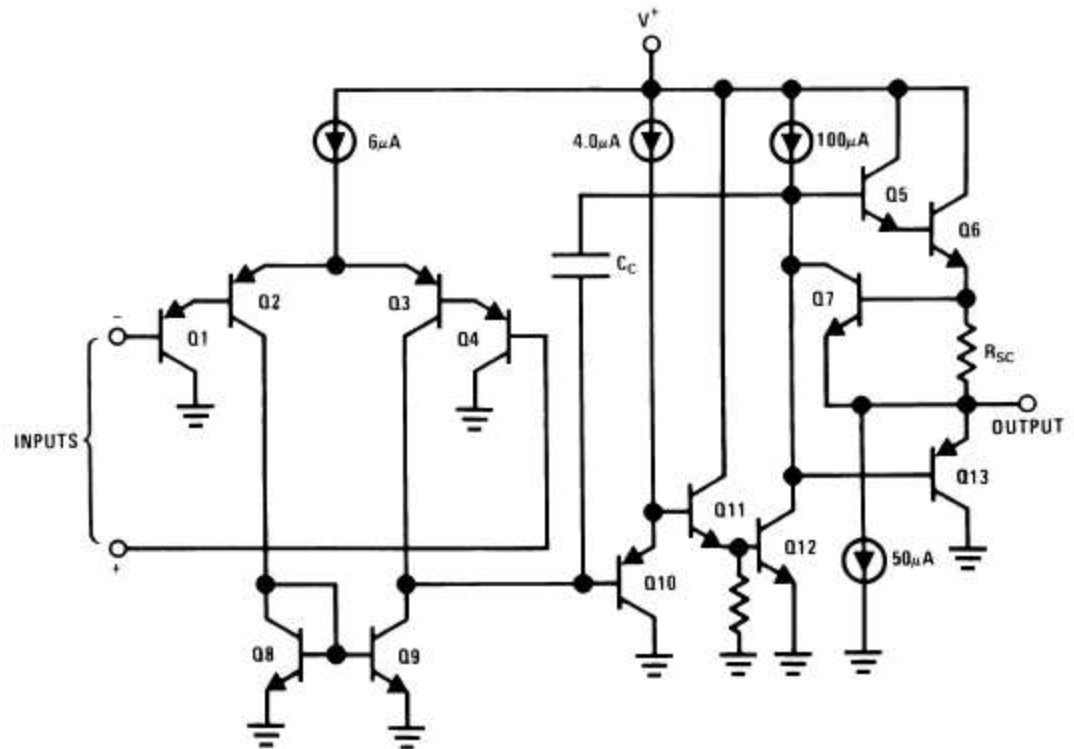


Top View

00778702

Advantages

- Two internally compensated op amps
- Eliminates need for dual supplies
- Allows direct sensing near GND and V_{OUT} also goes to GND
- Compatible with all forms of logic
- Power drain suitable for battery operation
- Pin-out same as LM1558/LM1458 dual op amp

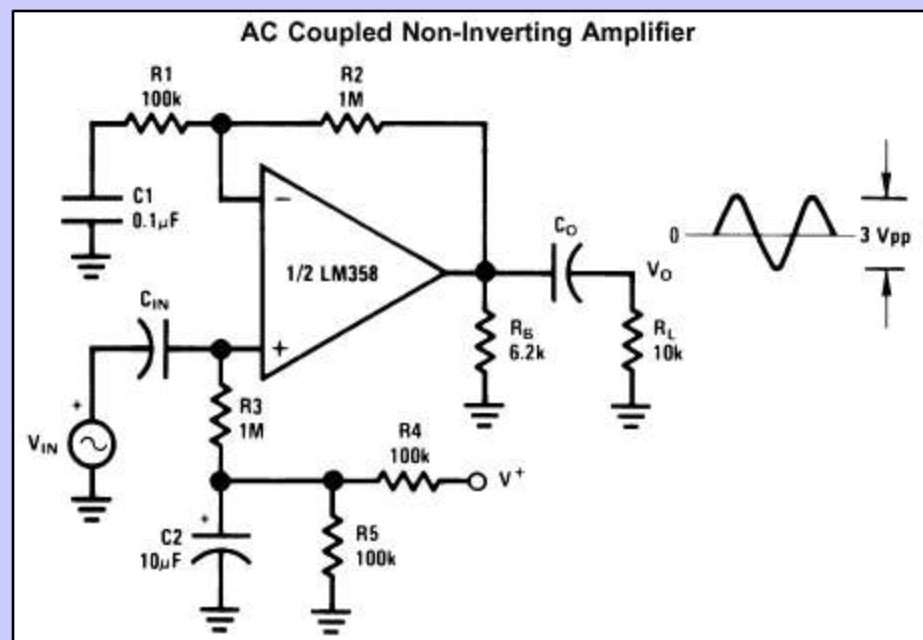
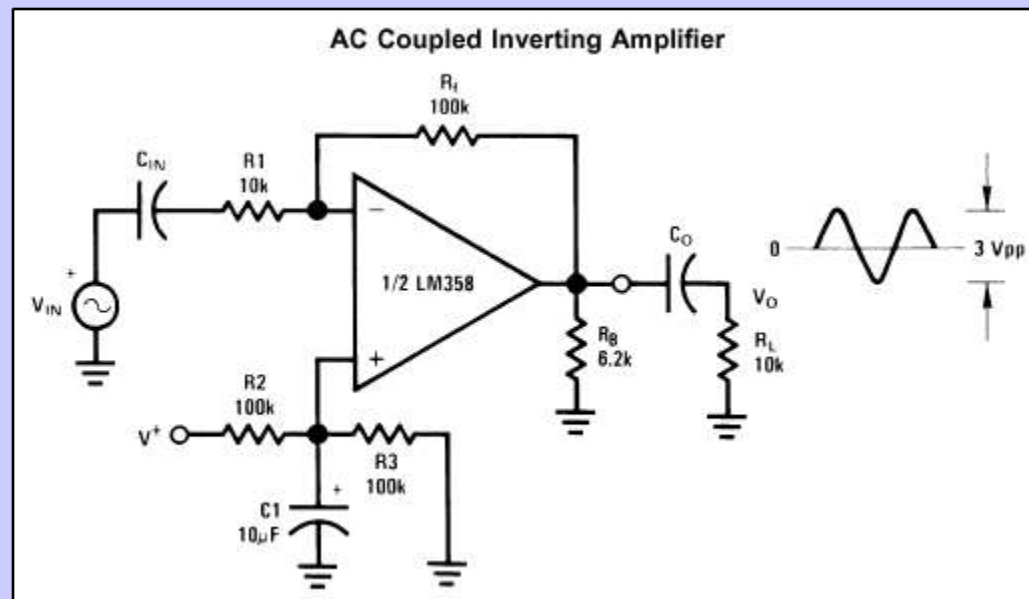


00778703

LM158/LM258/LM358/LM2904

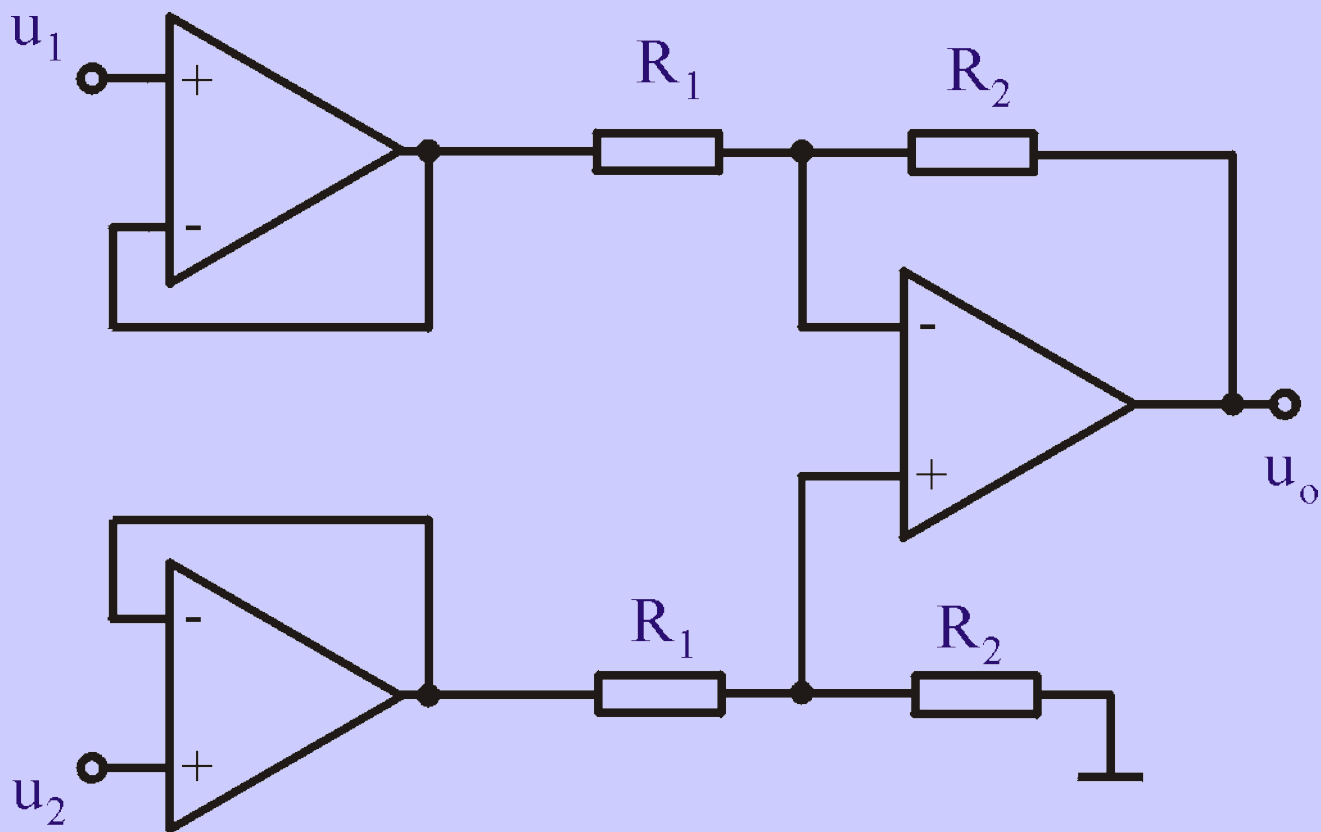
Low Power Dual Operational Amplifiers

Typical Single-Supply Applications ($V^+ = 5.0\text{ V}_{\text{DC}}$)



Instrumentförstärkare (förbättrad differentialförstärkare)

- hög ingångsimpedans, av alla krävs ett högt CMRR



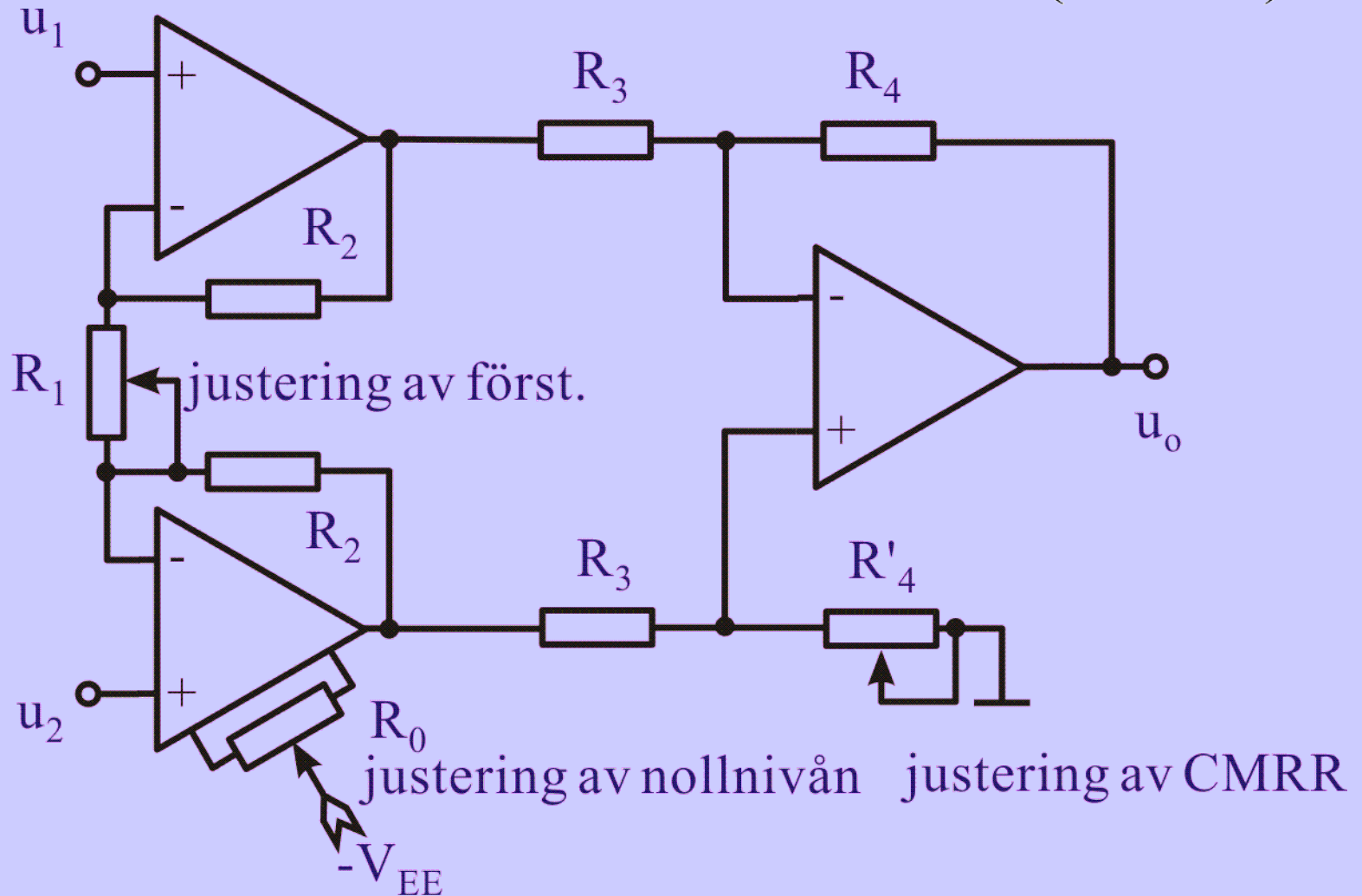
Mera info t.ex.

A Designer's Guide to Instrumentation Amplifiers

Typisk instrumentförstärkare

- de två första förstärkarna bör ha ett högt CMRR

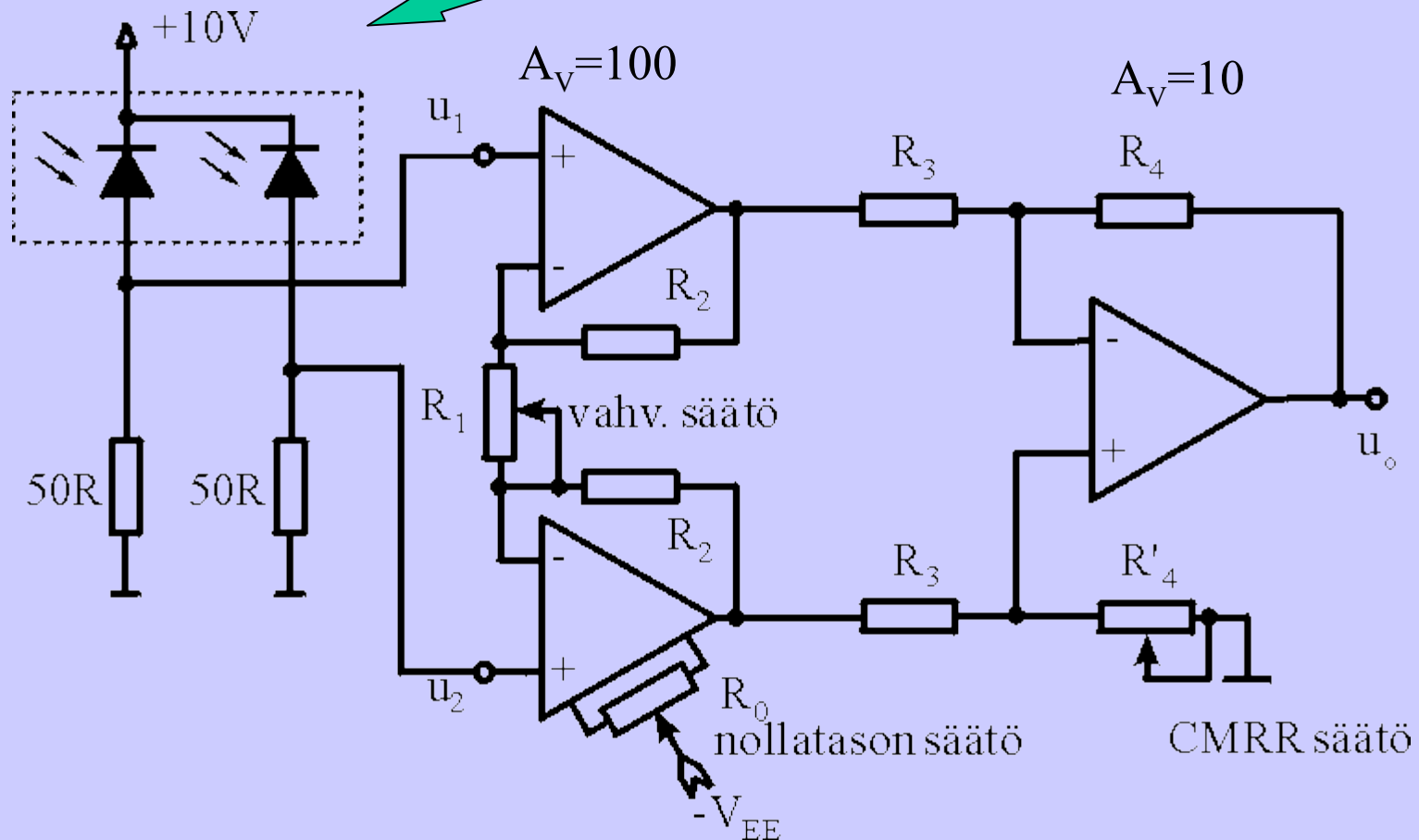
$$A_V = \left(1 + 2 \frac{R_2}{R_1} \right) \cdot \frac{R_4}{R_3}$$



exempel, uppskattning av bandbredd

totala förstärkningen $100 \cdot 10 = 1000$

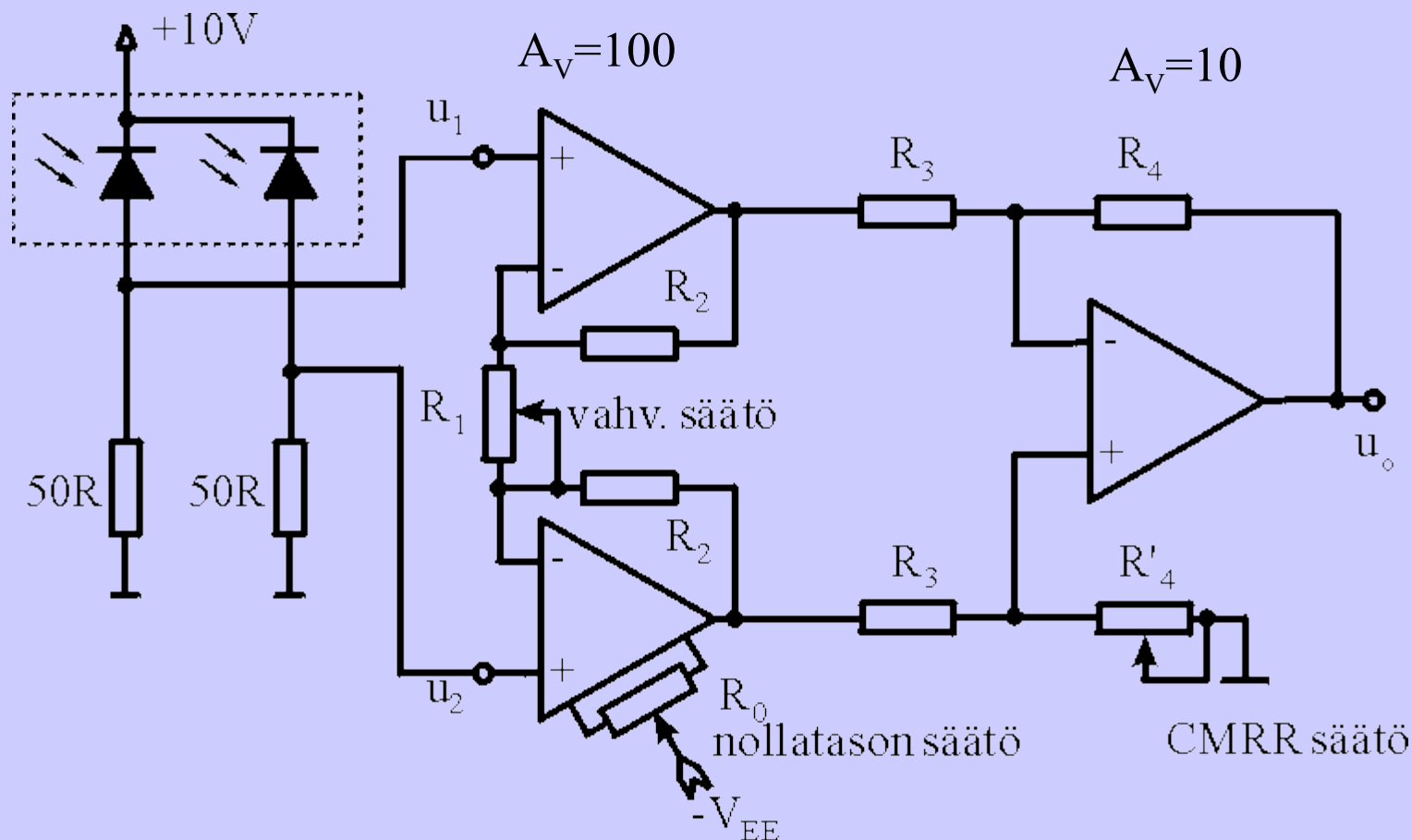
Hamamatsu S4204 dubbelfotodiodens bandbredd
är c. 30 MHz vid 50 ohms belastning



exempel, uppskattning av bandbredden

Ensimmäisen asteen vahvistimet ovat tyyppiä AD8001, vahvistimien $G \approx 100/2 = 50$, jolloin datalehtien mukaan kaistanleveys on $BW \approx 880\text{MHz}/50 = 18\text{ MHz}$

Toisen asteen vahvistin on tyyppiä AD8047, vahvistimen $G = 10$, jolloin datalehtien mukaan kaistanleveys on $BW \approx 250\text{MHz}/10 = 25\text{ MHz}$



exempel, uppsktning av bandbredden

AD8001—SPECIFICATIONS (@ T _A = + 25°C, V _S = ±5 V, R _L = 100 Ω, unless otherwise noted)						
Model	Conditions		AD8001A			Units
			Min	Typ	Max	
DYNAMIC PERFORMANCE						
-3 dB Small Signal Bandwidth,	N Package	G = +2, < 0.1 dB Peaking, R _F = 750 Ω	350	440		MHz
		G = +1, < 1 dB Peaking, R _F = 1 kΩ	650	880		MHz
	R Package	G = +2, < 0.1 dB Peaking, R _F = 681 Ω	350	440		MHz
		G = +1, < 0.1 dB Peaking, R _F = 845 Ω	575	715		MHz
	RT Package	G = +2, < 0.1 dB Peaking, R _F = 768 Ω	300	380		MHz
		G = +1, < 0.1 dB Peaking, R _F = 1 kΩ	575	795		MHz

AD8047/AD8048—SPECIFICATIONS								
ELECTRICAL CHARACTERISTICS ($\pm V_S = \pm 5\text{ V}$; $R_{\text{LOAD}} = 100\ \Omega$; $A_V = 1$ (AD8047); $A_V = 2$ (AD8048), unless otherwise noted)								
Parameter	Conditions	AD8047A			AD8048A			Units
		Min	Typ	Max	Min	Typ	Max	
DYNAMIC PERFORMANCE								
Bandwidth (-3 dB)								
Small Signal	$V_{\text{OUT}} \leq 0.4\text{ V p-p}$	170	250		180	260		MHz
Large Signal ¹	$V_{\text{OUT}} = 2\text{ V p-p}$	100	130		135	160		MHz
Bandwidth for 0.1 dB Flatness	$V_{\text{OUT}} = 300\text{ mV p-p}$							
	8047, $R_F = 0\ \Omega$; 8048, $R_F = 200\ \Omega$		35			50		MHz
Slew Rate, Average +/-	$V_{\text{OUT}} = 4\text{ V Step}$	475	750		740	1000		V/ μs
Rise/Fall Time	$V_{\text{OUT}} = 0.5\text{ V Step}$		1.1			1.2		ns
	$V_{\text{OUT}} = 4\text{ V Step}$		4.3			3.2		ns

exempel, uppskattning av bandbredd

AD8001

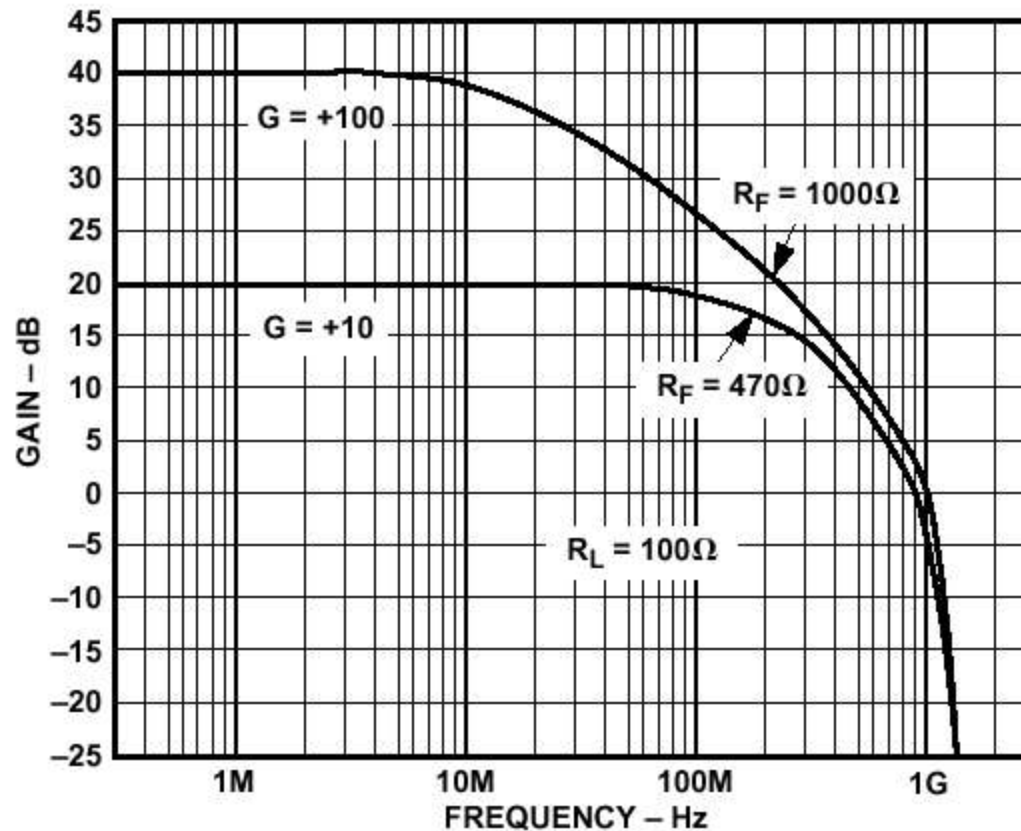
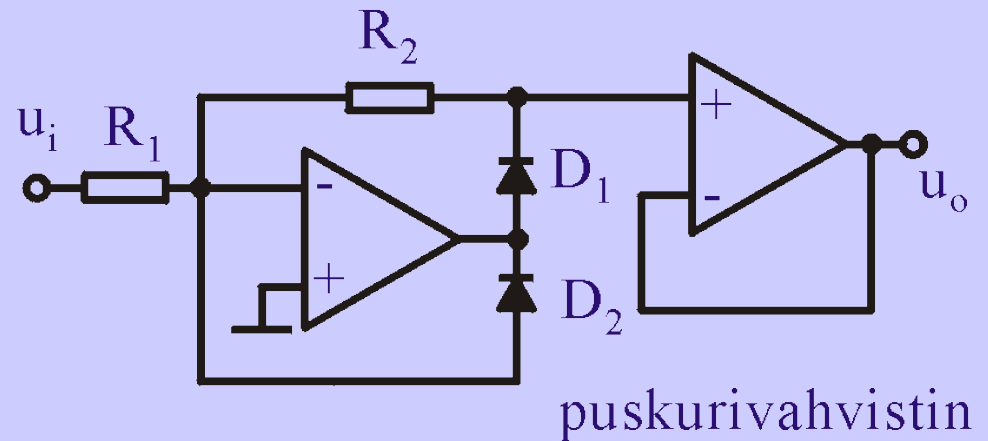
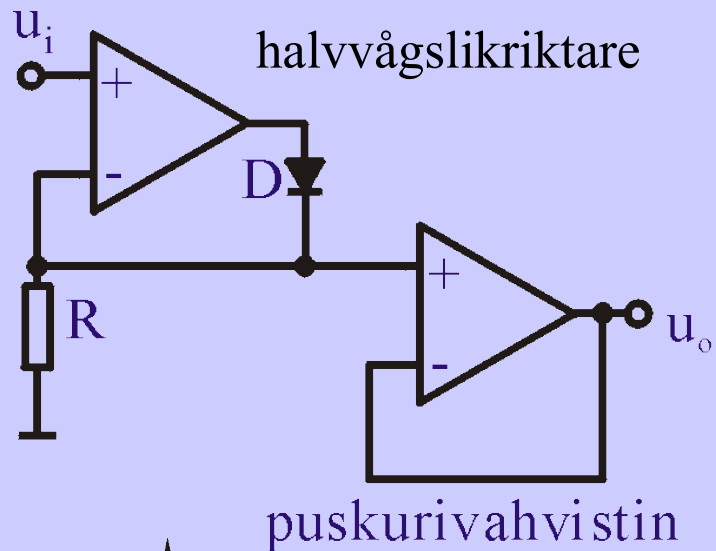
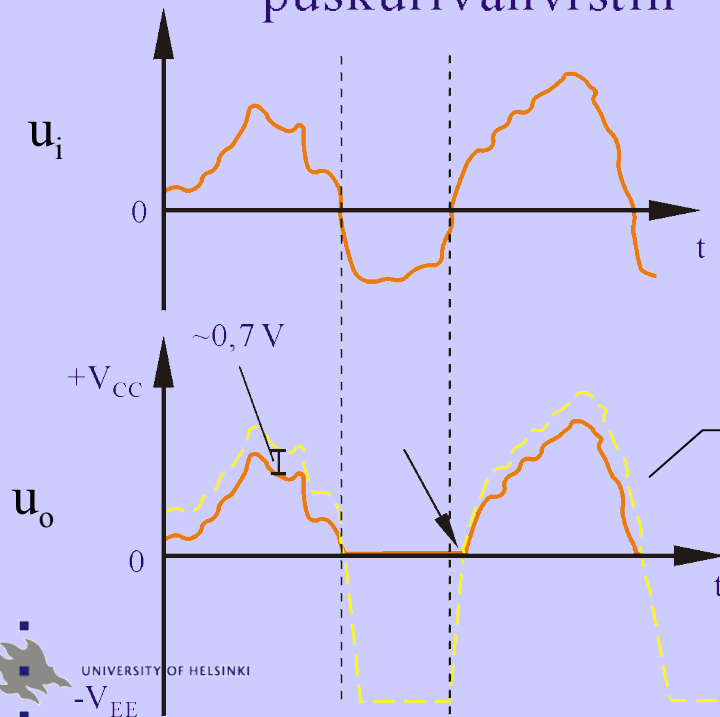


Figure 22. Frequency Response, $G = +10$, $G = +100$

Aktiva likriktare

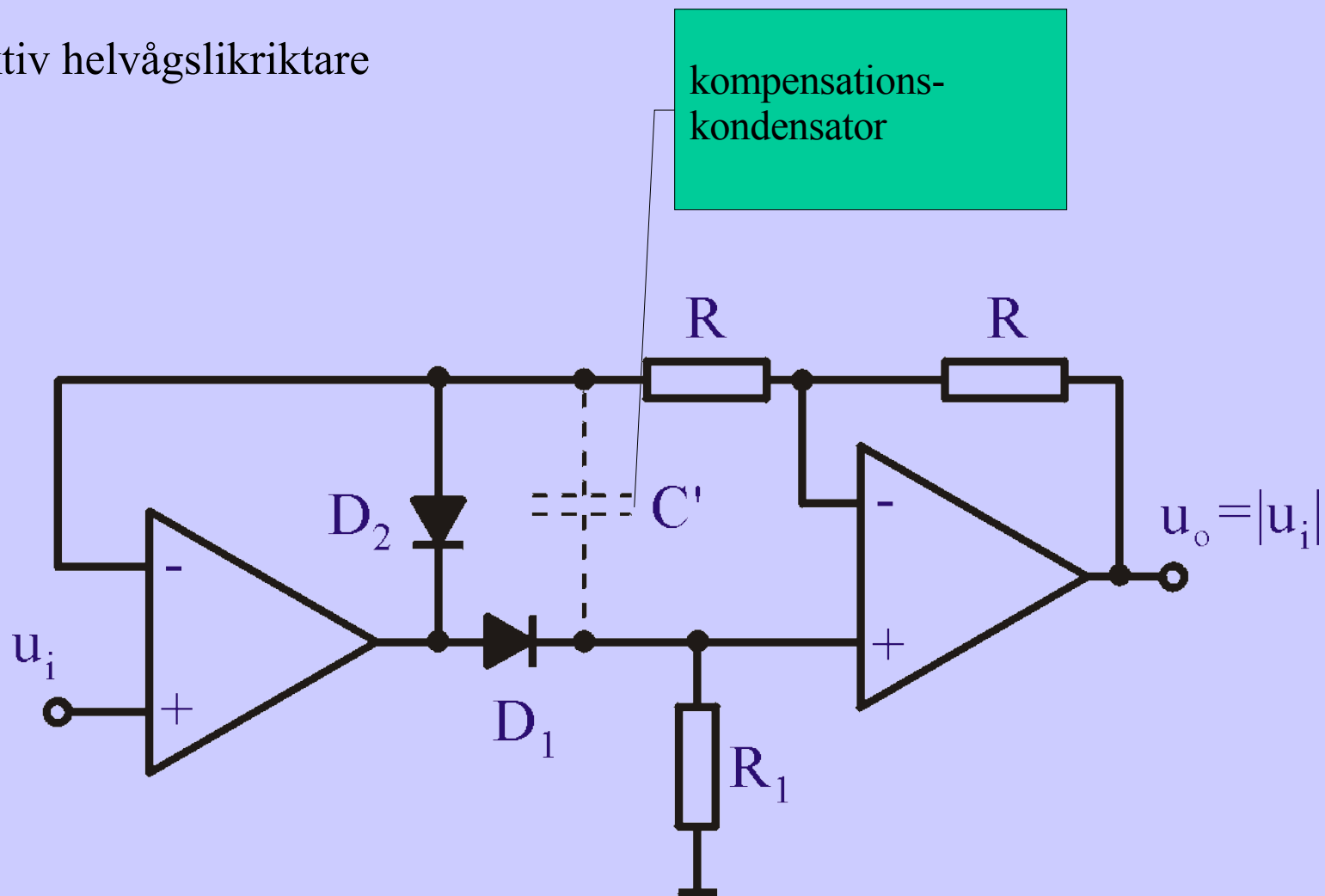


Halvvågsl rikriktare som fungerar bättre för snabba signaler

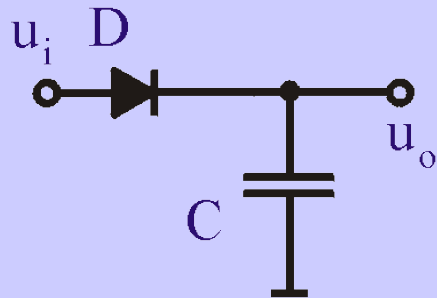


1a operationsförtärkarens utgångsspänning

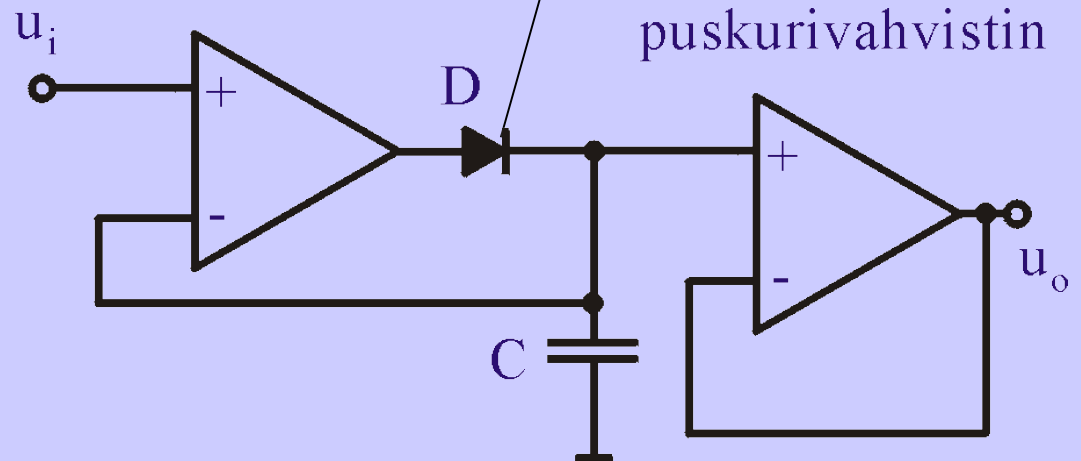
Aktiv halvågslikriktare



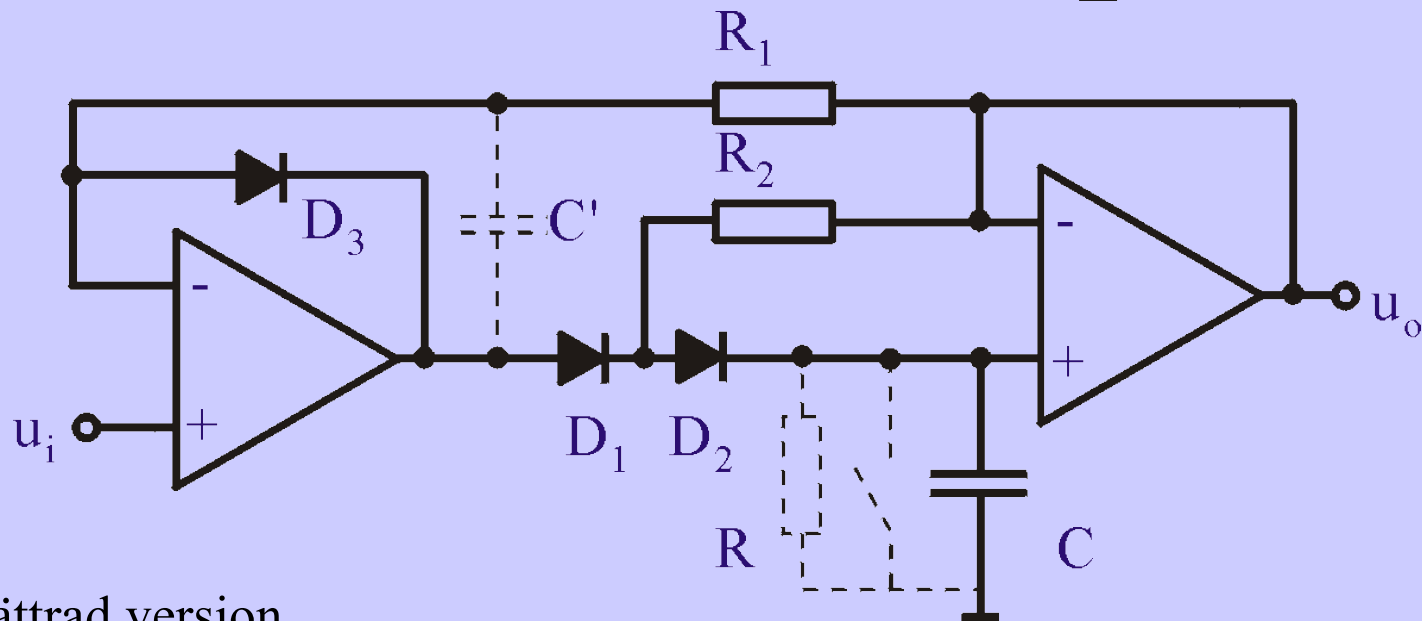
Topvärdesdetektor (*peak detector*)



enkel

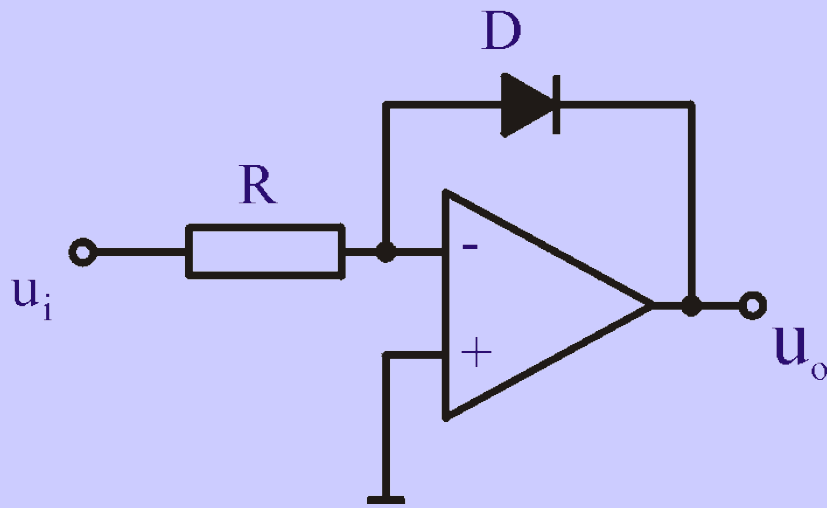


puskurivahvistin

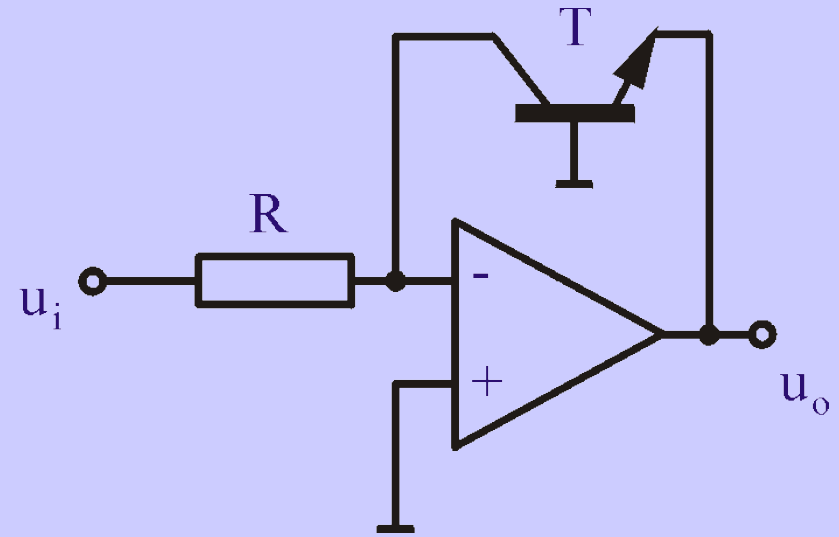


förbättrad version

Logaritmisk förstärkare



$$u_D = u_o \approx -2,3 \frac{kT}{q} \log\left(\frac{i_D}{I_S}\right)$$



$$u_o = V_E \approx -2,3 \frac{kT}{q} \log\left(\frac{i_C}{I_S}\right)$$

- vid signalbehandling t.ex. för att öka på dynamiken
- analog multiplikation och division

Multiplikation:

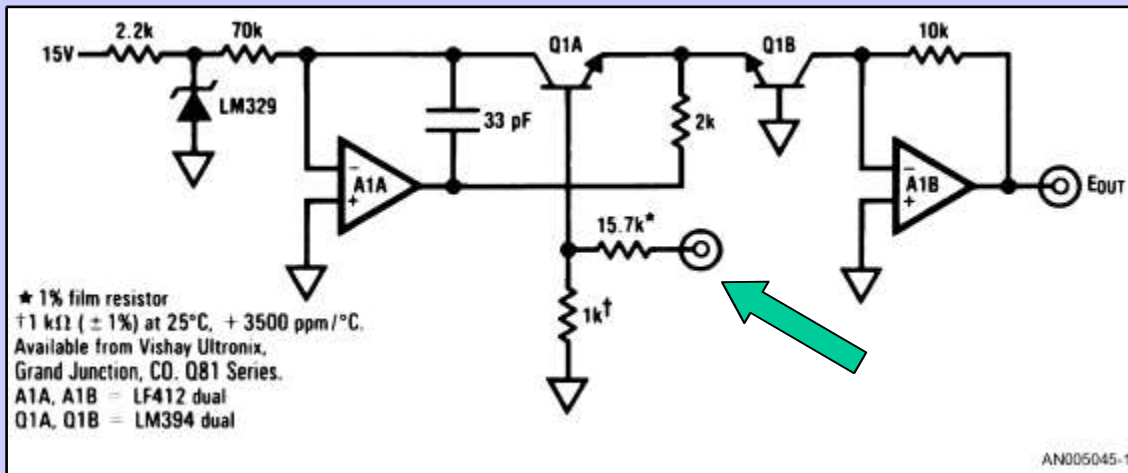
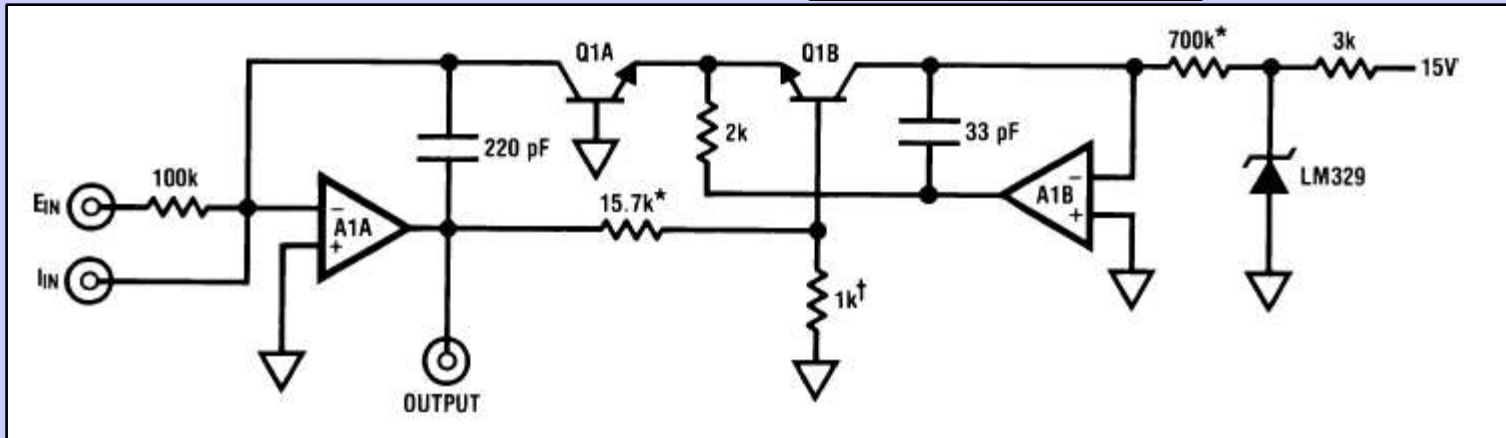
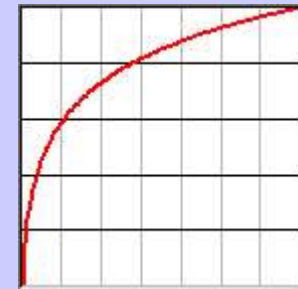
$$\ln a + \ln b = \ln ab$$

$$e^{\ln ab} = ab$$

Division:

$$\ln a - \ln b = \ln \frac{a}{b}$$

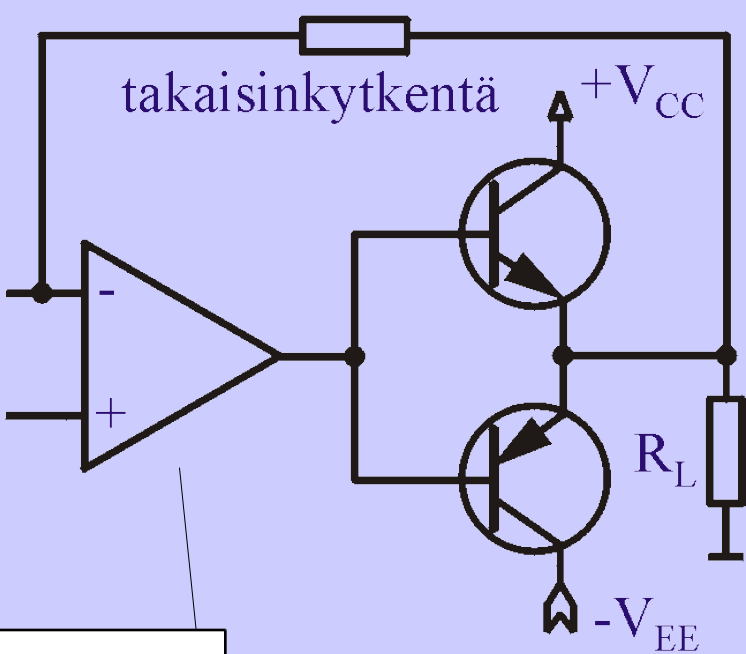
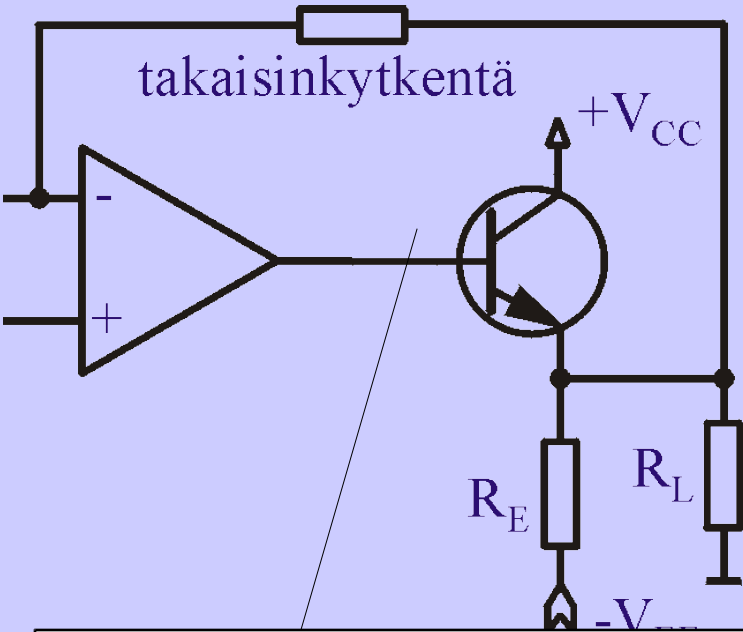
$$e^{\ln \frac{a}{b}} = \frac{a}{b}$$



National semiconductor



Ulostulon virrankesto



LM158/LM258/LM358/LM2904

Output Current	Source	$V_{IN}^+ = 1V$					
----------------	--------	-----------------	--	--	--	--	--

Parameter	Conditions	AD8047A			AD8048A			Units
		Min	Typ	Max	Min	Typ	Max	
OUTPUT CHARACTERISTICS								
Output Voltage Range, R _L = 150 Ω		±2.8	±3.0		±2.8	±3.0		V
Output Current			50			50		mA
Output Resistance			0.2			0.2		Ω
Short Circuit Current			130			130		mA

Short Circuit to Ground	$V^- = 15V$ $T_A = 25^\circ C$, (Note 2), $V^+ = 15V$	40	60	40	60	40	60	mA
-------------------------	--	----	----	----	----	----	----	----

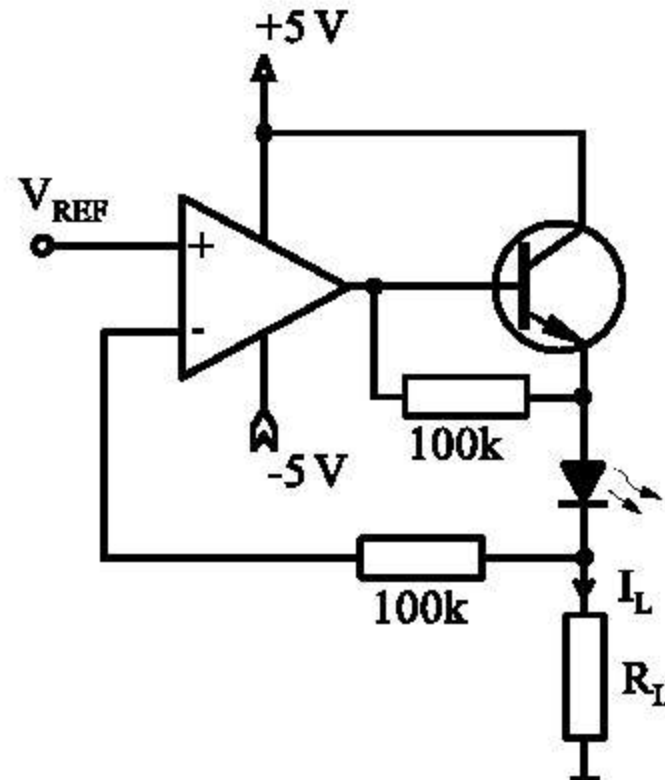
Esimerkki 5.1

Valodiodin lävitse halutaan kulkevan vakiovirta. Mikä on virta I_L (ja samalla valodiodin virta) seuraavassa kytkennässä, kun $V_{REF} = 0,6 \text{ V}$ ja $R_L = 30 \Omega$?

Operaatiovahvistimen sisäänmenojen jännite-ero pyrkii olemaan nolla ja sen sisäänmenoihin ei kulje virtaa, joten tästä seuraa, että kuormavastuksen R_L yli on jännite:

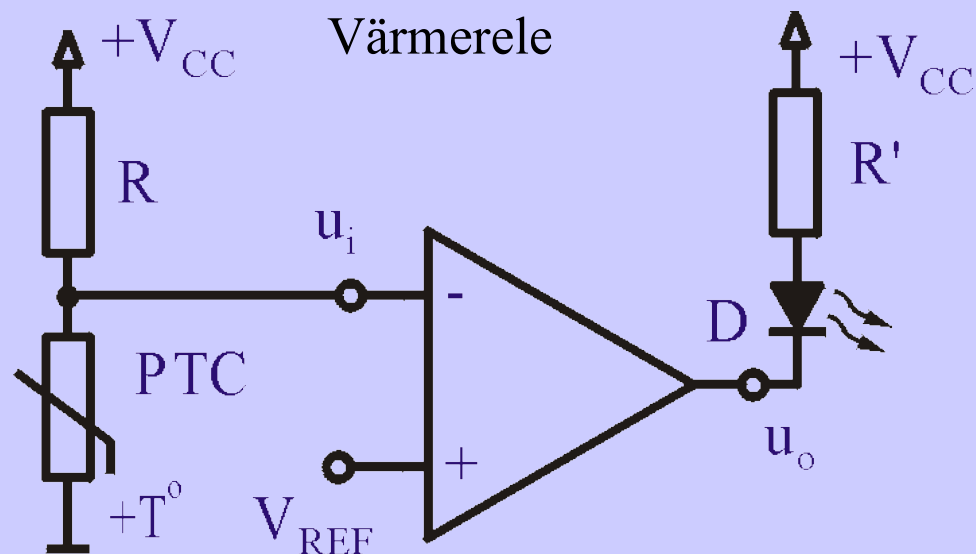
$U_{RL} = V_{REF} = 0,6 \text{ V}$ ja virta

$$I_{\text{valodiodi}} = I_L = \frac{U_{RL}}{R_L} = 20 \text{ mA} .$$

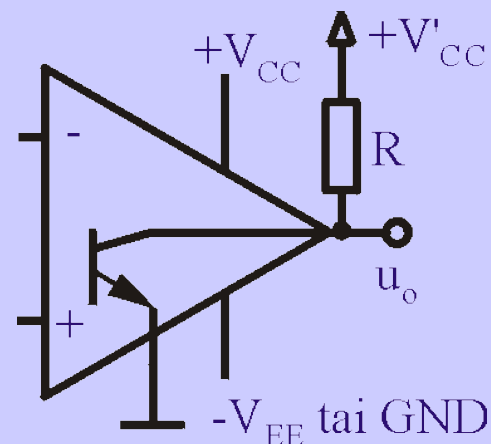


Komparator

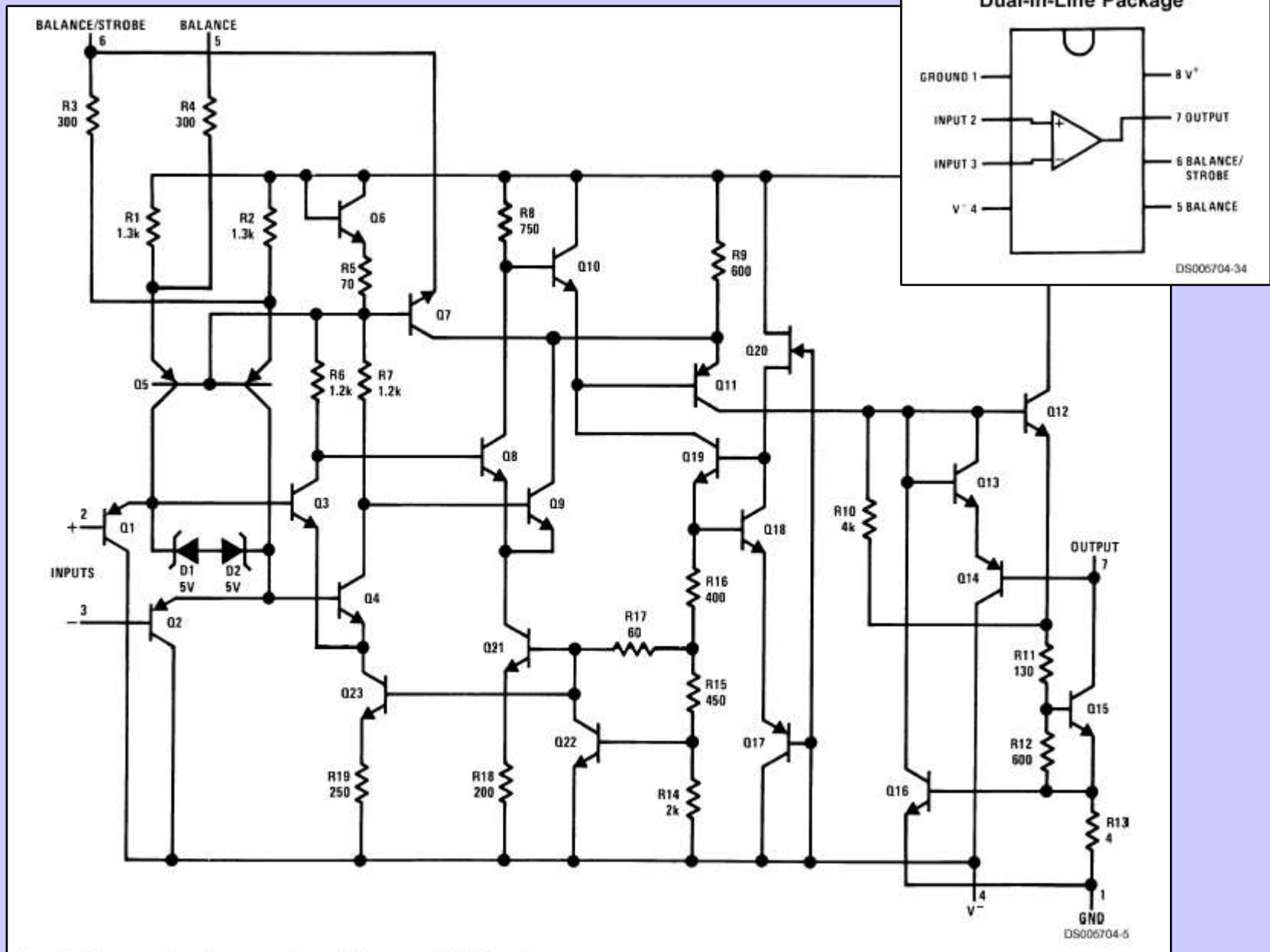
- icke tillbakakopplad operationsförstärkare



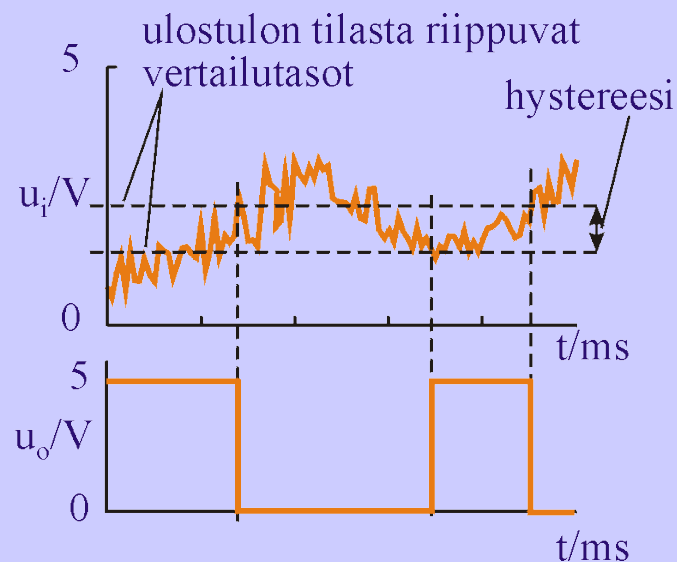
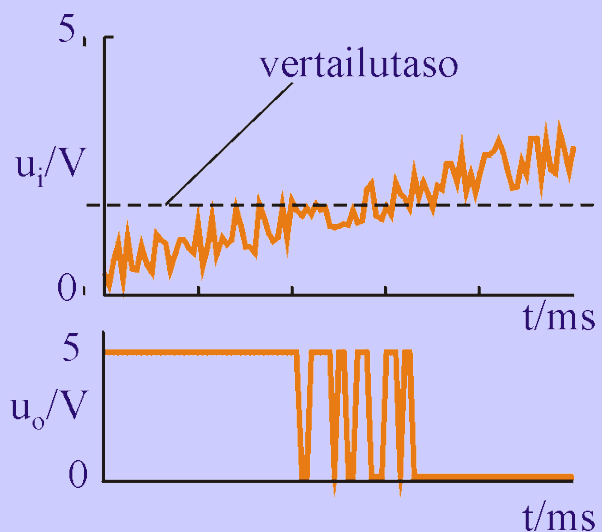
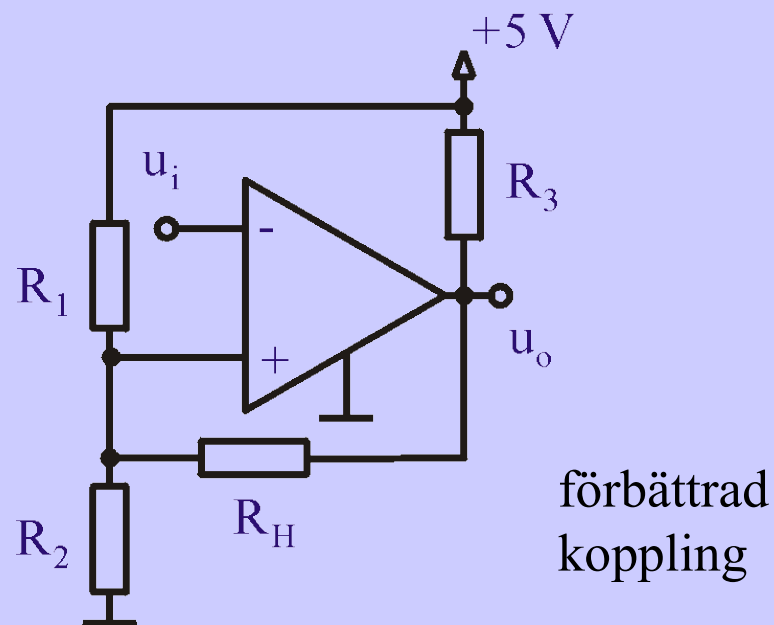
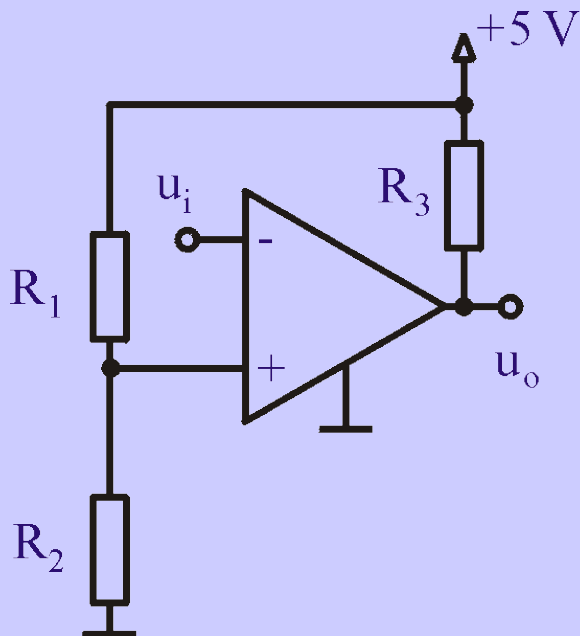
- komparatorns utgång är ofta en öppen kollektor



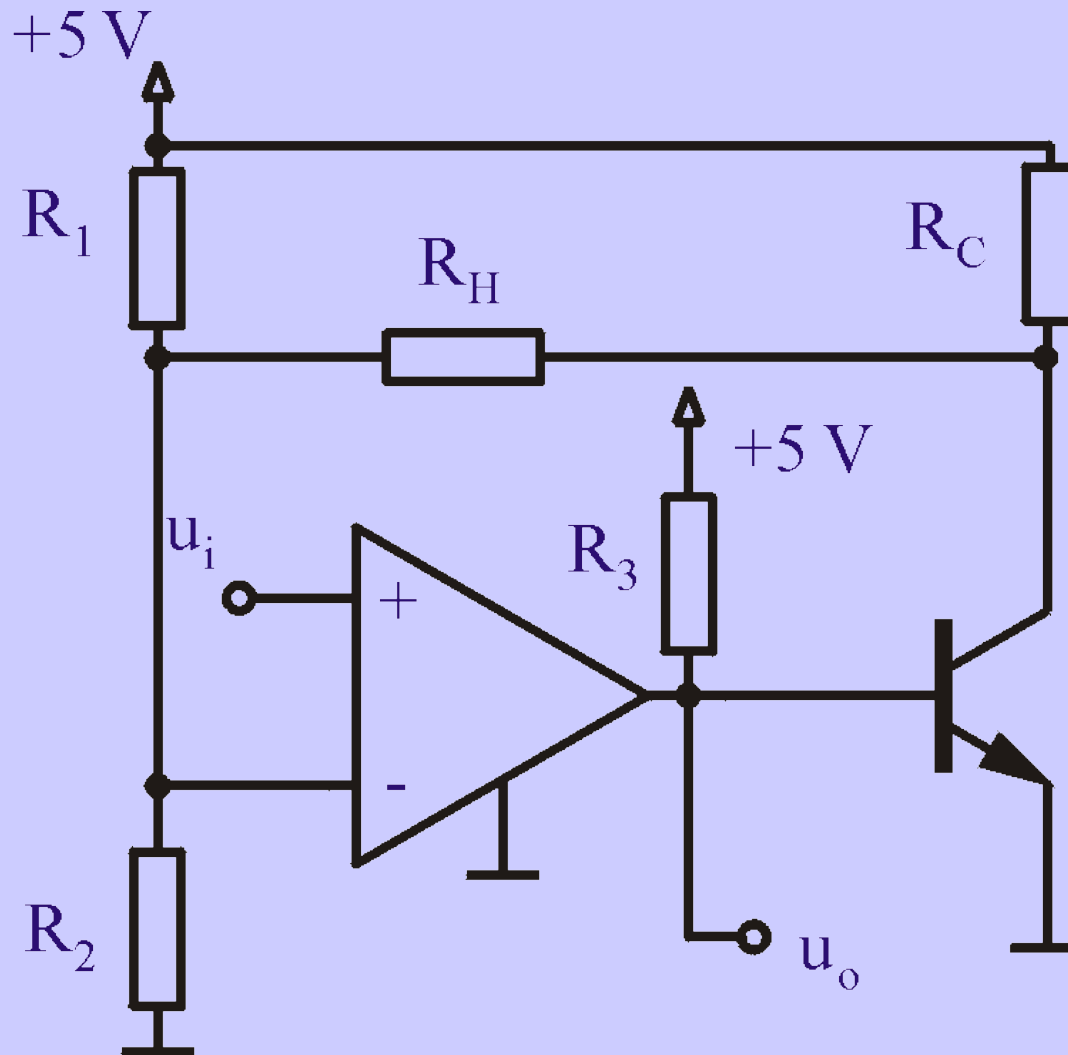
LM311



Komparatorkopplingar

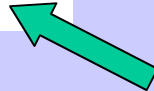


Icke-inverterande komparatorkoppling



LM311, ohjeita datalehdeltä

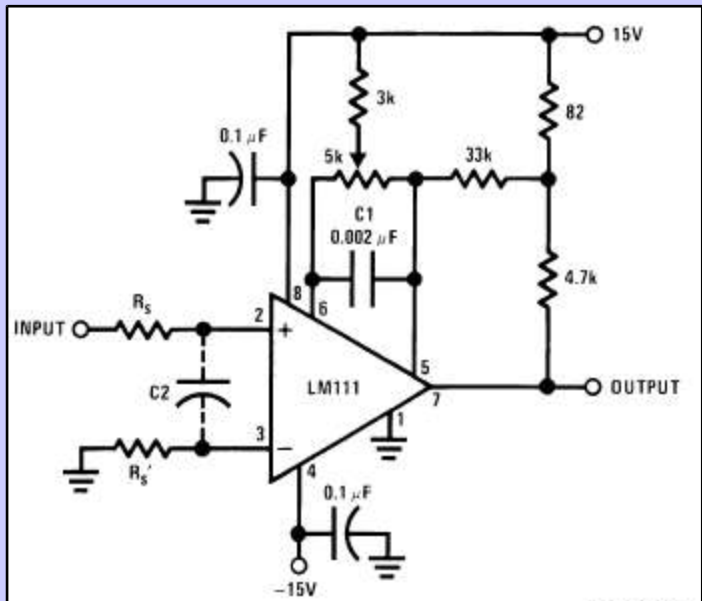
3. When the signal source is applied through a resistive network, R_S , it is usually advantageous to choose an R_S of substantially the same value, both for DC and for dynamic (AC) considerations. Carbon, tin-oxide, and metal-film resistors have all been used successfully in comparator input circuitry. Inductive wirewound resistors are not suitable.



4. When comparator circuits use input resistors (eg. summing resistors), their value and placement are particularly important. In all cases the body of the resistor should be close to the device or socket. In other words there should be very little lead length or printed-circuit foil run between comparator and resistor to radiate or pick up signals. The same applies to capacitors, pots, etc. For example, if $R_S = 10 \text{ k}\Omega$, as little as 5 inches of lead between the resistors and the input pins can result in oscillations that are very hard to damp. Twisting these input leads tightly is the only (second best) alternative to placing resistors close to the comparator.

5. Since feedback to almost any pin of a comparator can result in oscillation, the printed-circuit layout should be engineered thoughtfully. Preferably there should be a groundplane under the LM111 circuitry, for example, one side of a double-layer circuit card. Ground foil (or, positive supply or negative supply foil) should extend between the output and the inputs, to act as a guard. The foil connections for the inputs should be as small and compact as possible, and should be essentially surrounded by ground foil on all sides, to guard against capacitive coupling from any high-level signals (such as the output). If pins 5 and 6 are not used, they should be shorted together. If they are connected to a trim-pot, the trim-pot should be located, at most, a few inches away from the LM111, and the $0.01 \mu\text{F}$ capacitor should be installed. If this capacitor cannot be used, a shielding printed-circuit foil may be advisable between pins 6 and 7. The power supply bypass capacitors should be located within a couple inches of the LM111. (Some other comparators require the power-supply bypass to be located immediately adjacent to the comparator.)

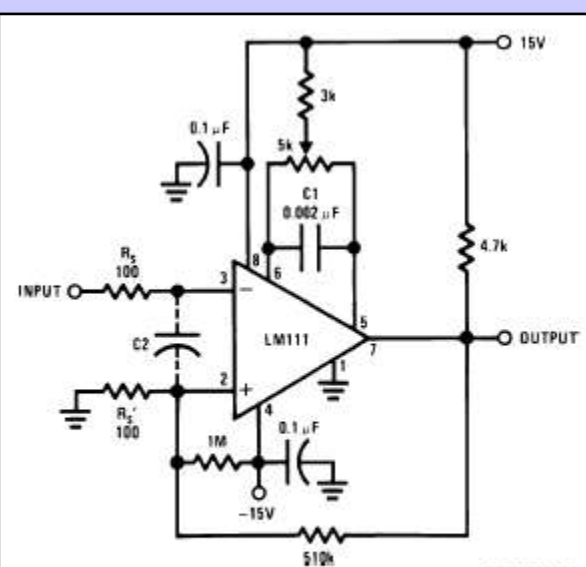
6. It is a standard procedure to use hysteresis (positive feedback) around a comparator, to prevent oscillation, and to avoid excessive noise on the output because the comparator is a good amplifier for its own noise. In the circuit of *Figure 2*, the feedback from the output to the positive input will cause about 3 mV of hysteresis. However, if R_S is larger than 100Ω , such as $50 \text{ k}\Omega$, it would not be reasonable to simply increase the value of the positive feedback resistor above $510 \text{ k}\Omega$. The circuit of *Figure 3* could be used, but it is rather awkward. See the notes in paragraph 7 below.



DS005704-29

the H08 hermetic package

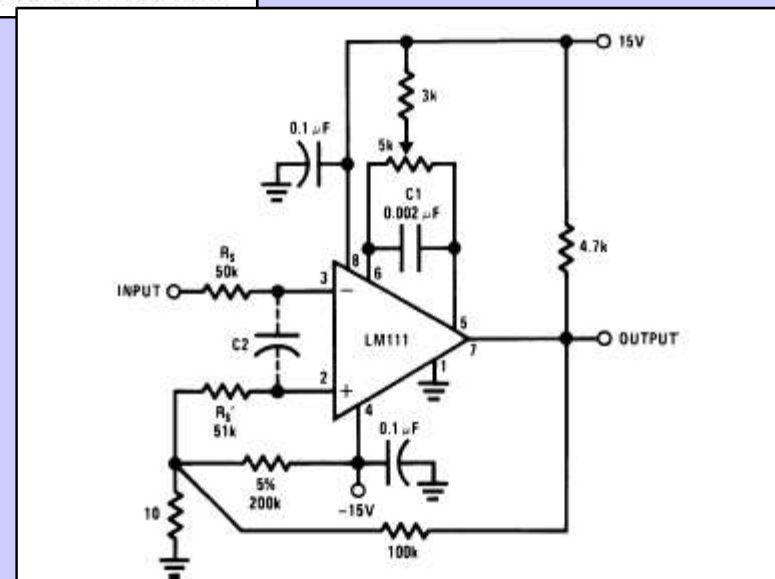
FIGURE 1. Improved Positive Feedback



DS005704-30

08 hermetic package

FIGURE 2. Conventional Positive Feedback



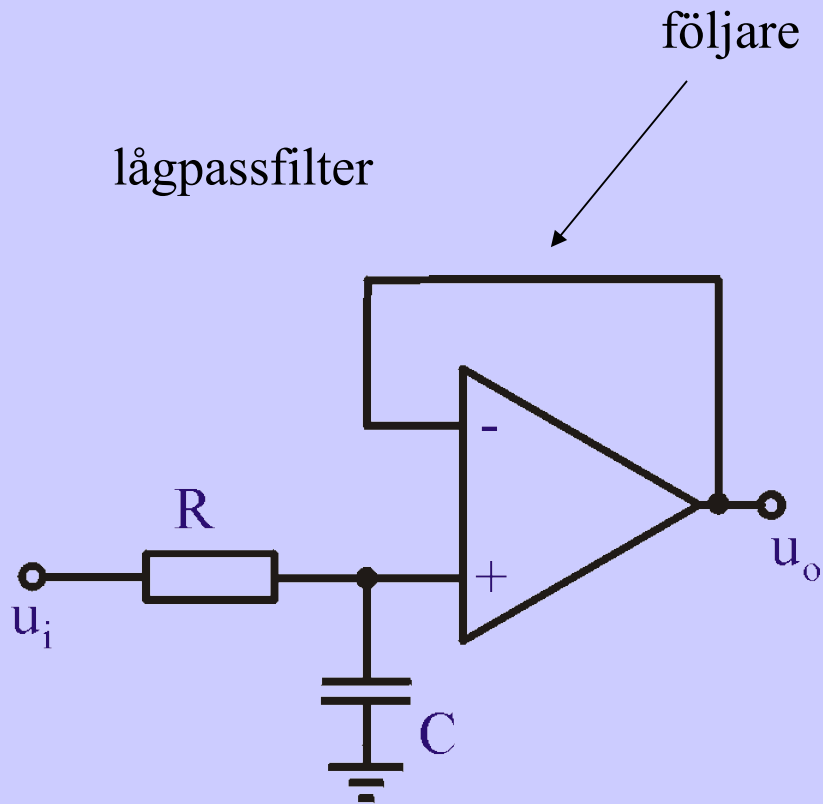
DS005704-31

FIGURE 3. Positive Feedback with High Source Resistance

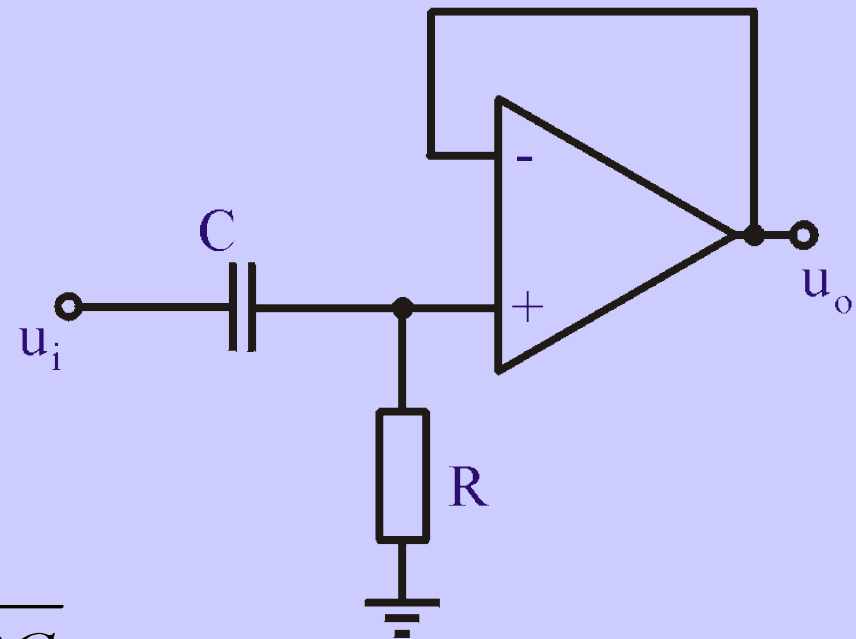
Aktiva filter

1:a ordningens

lågpassfilter



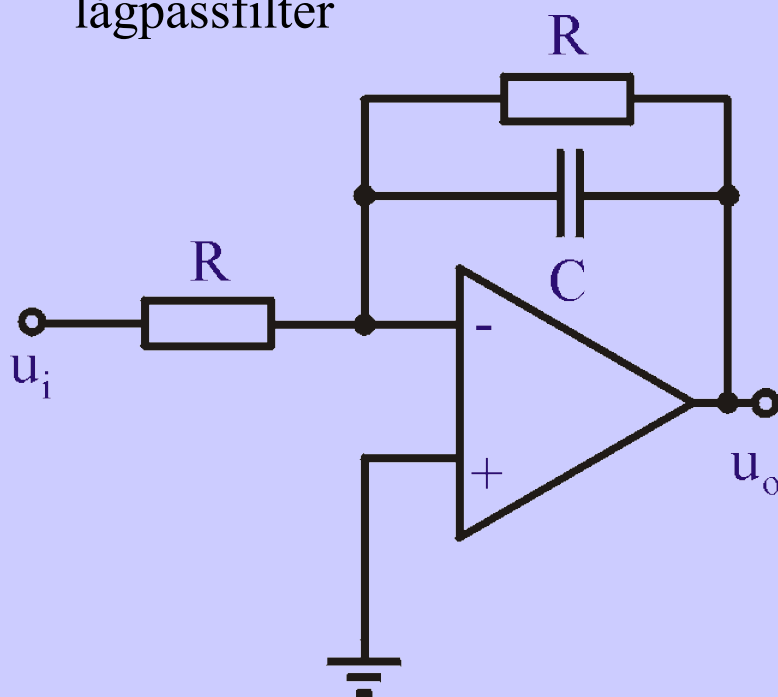
högpassfilter



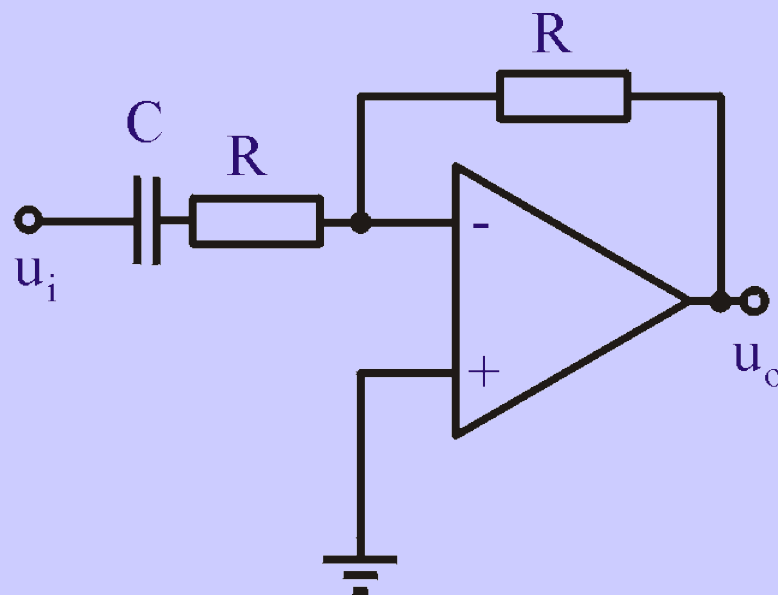
$$f_c = \frac{1}{2\pi RC}$$

Inverterande 1:a ordningens

lågpassfilter



högpassfilter



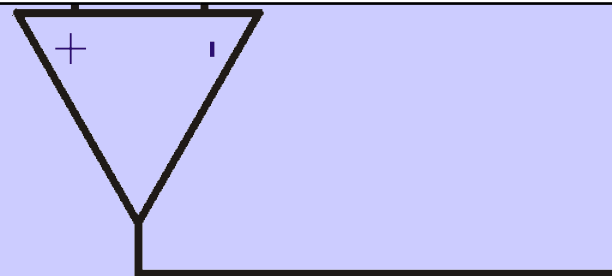
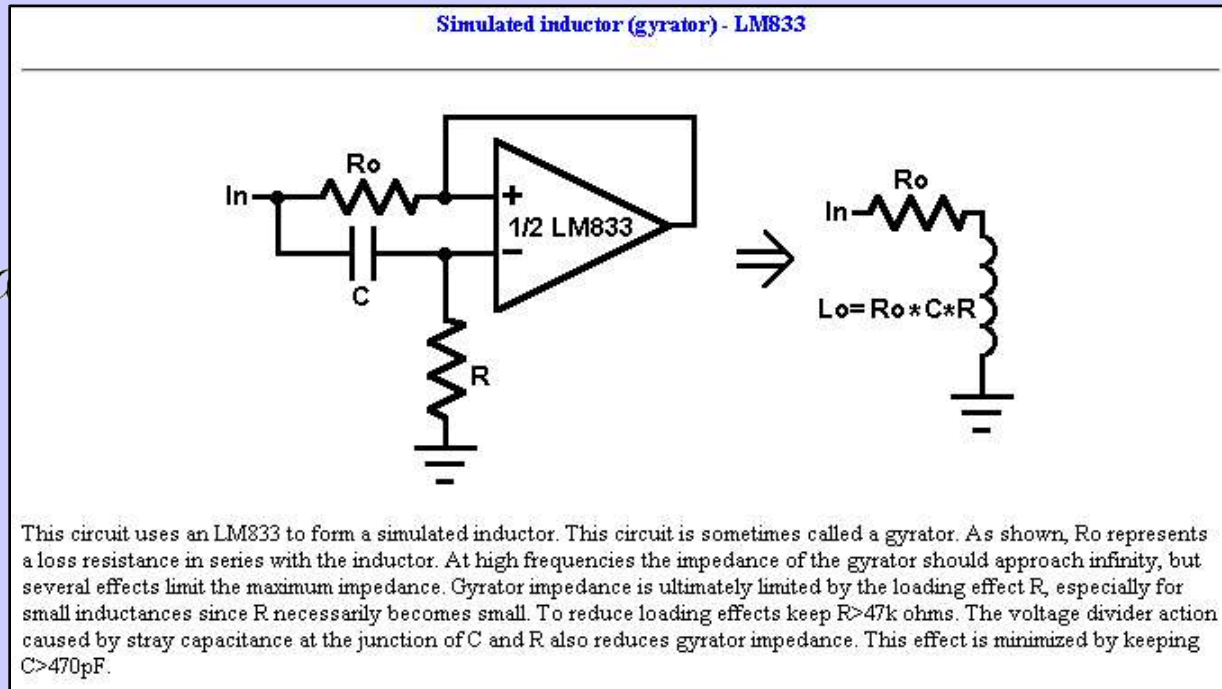
$$f_c = \frac{1}{2\pi RC}$$

Gyraattori

- kelamainen toiminta operaatiovahvistimien avulla toteutettuna

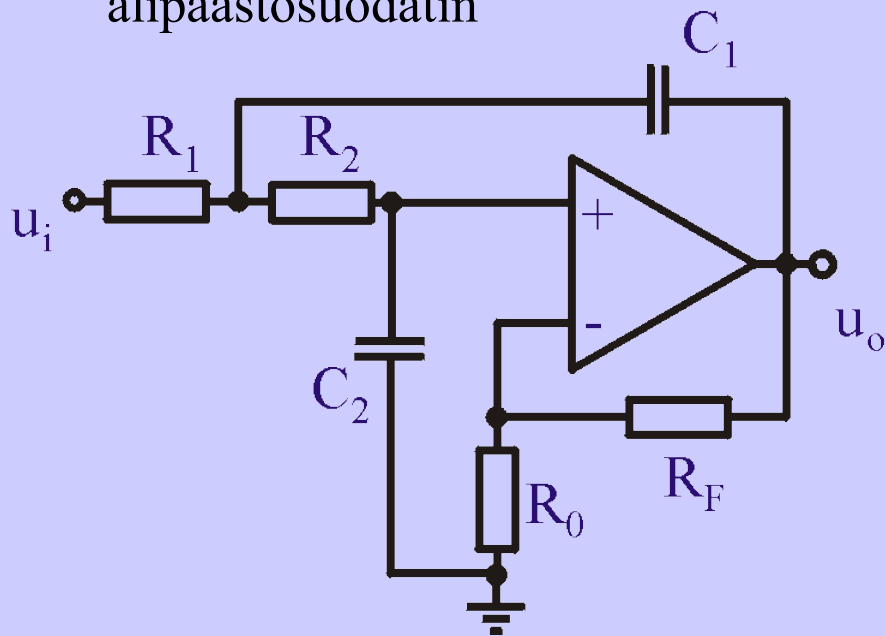
$$Z_i = j\omega L$$

Z_i

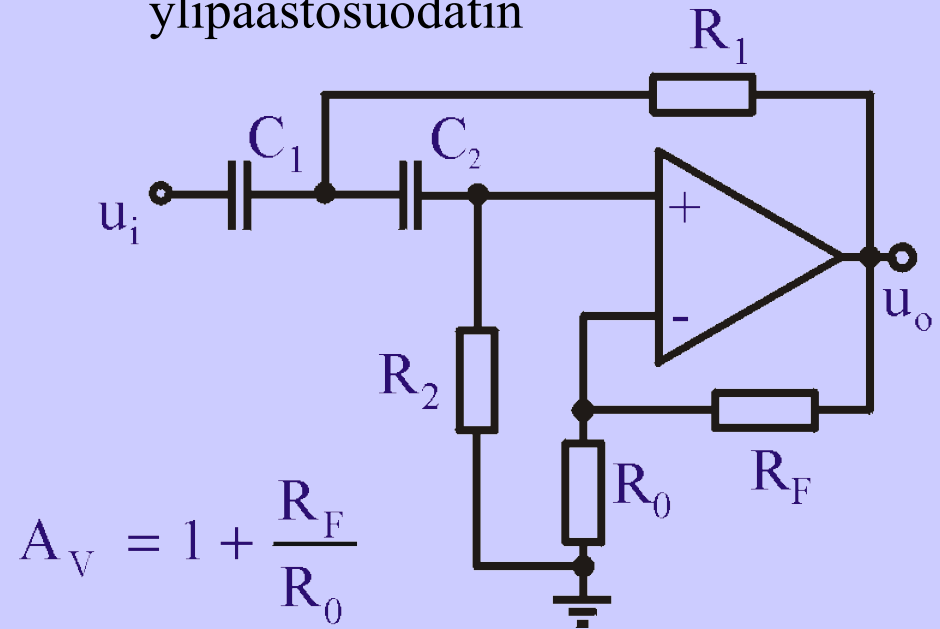


2. kertaluvun

alipäästösuodatin



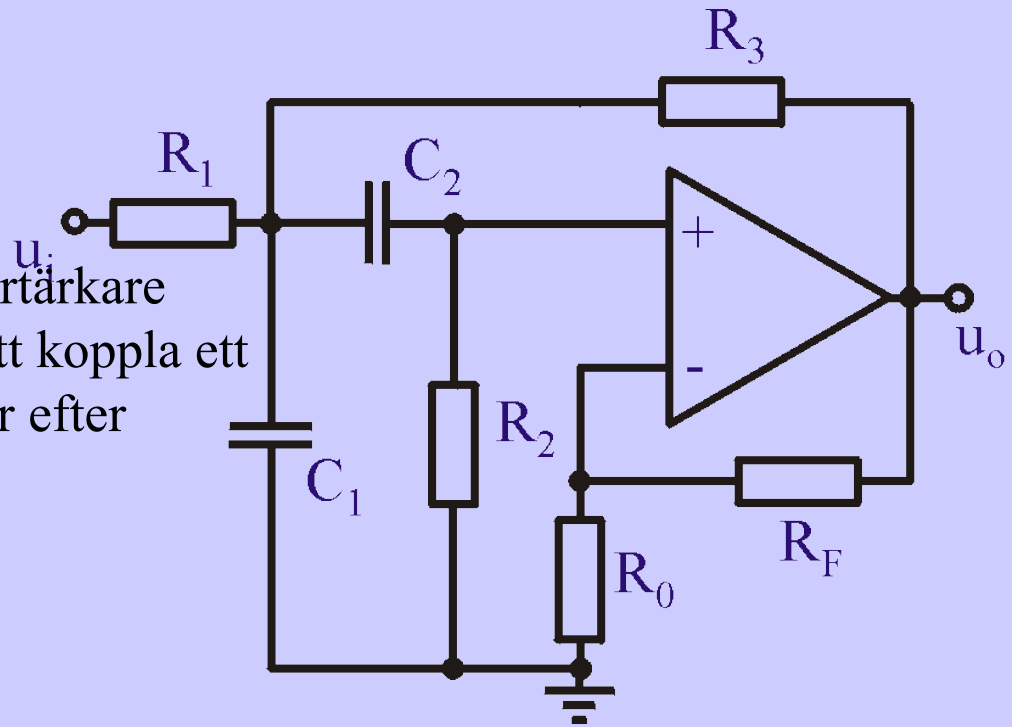
ylipäästösuodatin



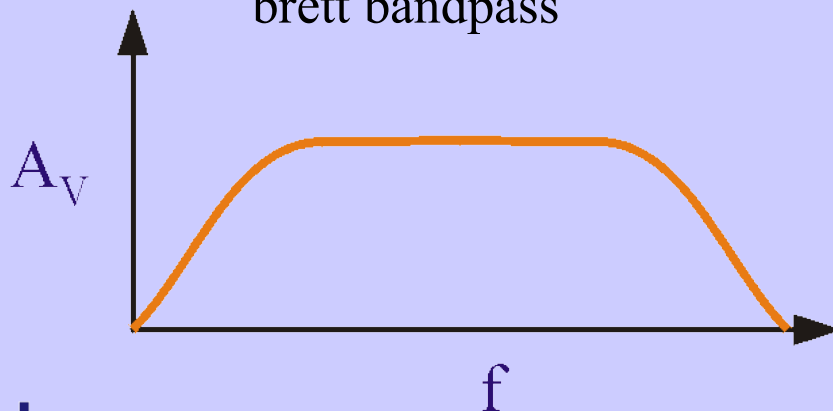
$$A_V = 1 + \frac{R_F}{R_0}$$

Sallen-Keyn kaistanpäästösuodatin

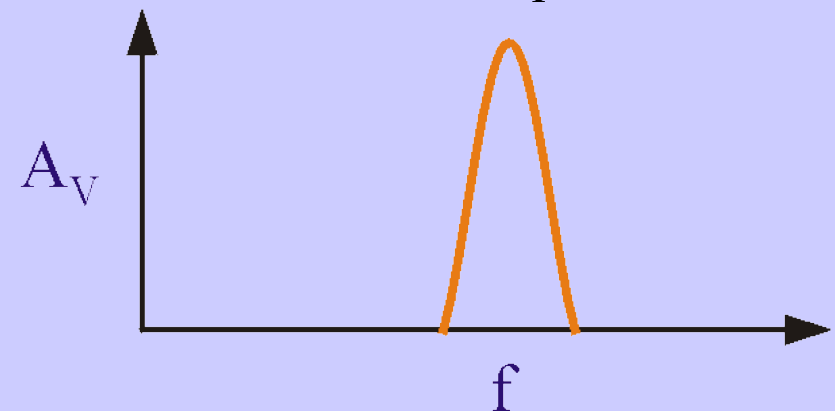
- ett smalt bandpassfilter kan förverkligas med en operationsförtärkare
- ett brett bandpassfilter fås med att koppla ett lågpass- och ett högpasfilter efter varandra



brett bandpass



smalt bandpass



kertaluku	Butterworth	Bessel		Chebyshev 0,5 dB		Chebyshev 2 dB	
	A_V	f_n	A_V	f_n	A_V	f_n	A_V
2	1,586	1,272	1,268	1,231	1,842	0,907	2,114
4	1,152	1,432	1,084	0,597	1,582	0,471	1,924
	2,235	1,606	1,759	1,031	2,660	0,964	2,782
6	1,068	1,607	1,040	0,396	1,537	0,316	1,891
	1,586	1,692	1,364	0,768	2,448	0,730	2,648
	2,483	1,908	2,023	1,011	2,846	0,983	2,904
8	1,038	1,781	1,024	0,297	1,522	0,238	1,879
	1,337	1,835	1,213	0,599	2,379	0,572	2,605
	1,889	1,956	1,593	0,861	2,711	0,842	2,821
	2,610	2,192	2,184	1,006	2,913	0,990	2,946

Taulukko 5.1. Kolmen suodatintyyppin suunnitteluparametrit (ks. esimerkki 5.2).



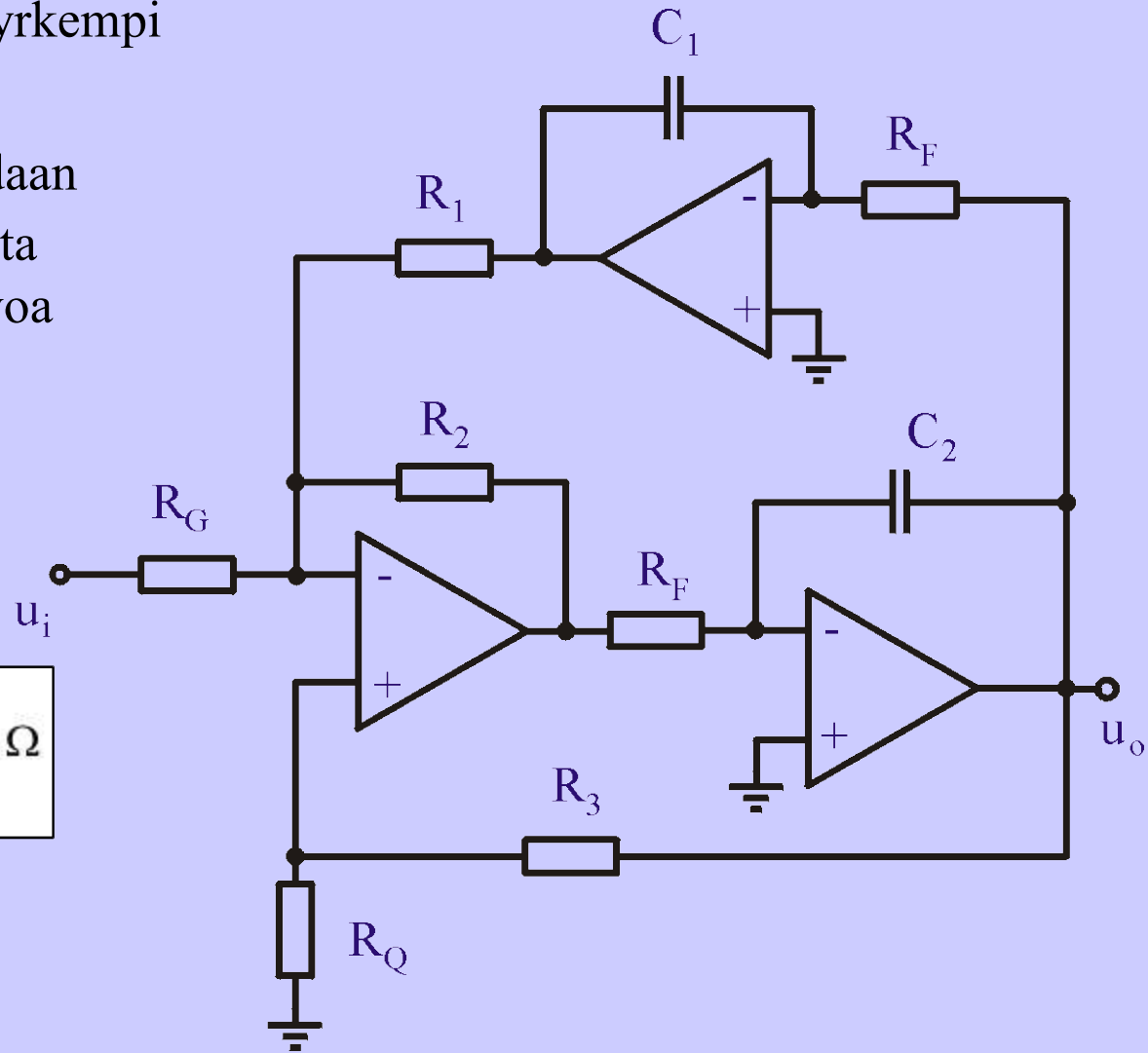
Kolmen operaatiovahvistimen kaistanpäästösuodatin (*state-variable filter*)

- saadaan edellistä jyrkempi kaistanpäästö
- vastuksilla R_F voidaan säätää keskitaajuutta muuttamatta Q -arvoa

$$R_F = \frac{5,03 \cdot 10^7 \frac{\Omega}{s}}{f_0} ,$$

$$R_Q = \frac{10^5}{(3,48 \cdot Q + A_V - 1)} \Omega$$

$$R_G = \frac{3,16 \cdot 10^4 \cdot Q}{A_V} \Omega .$$



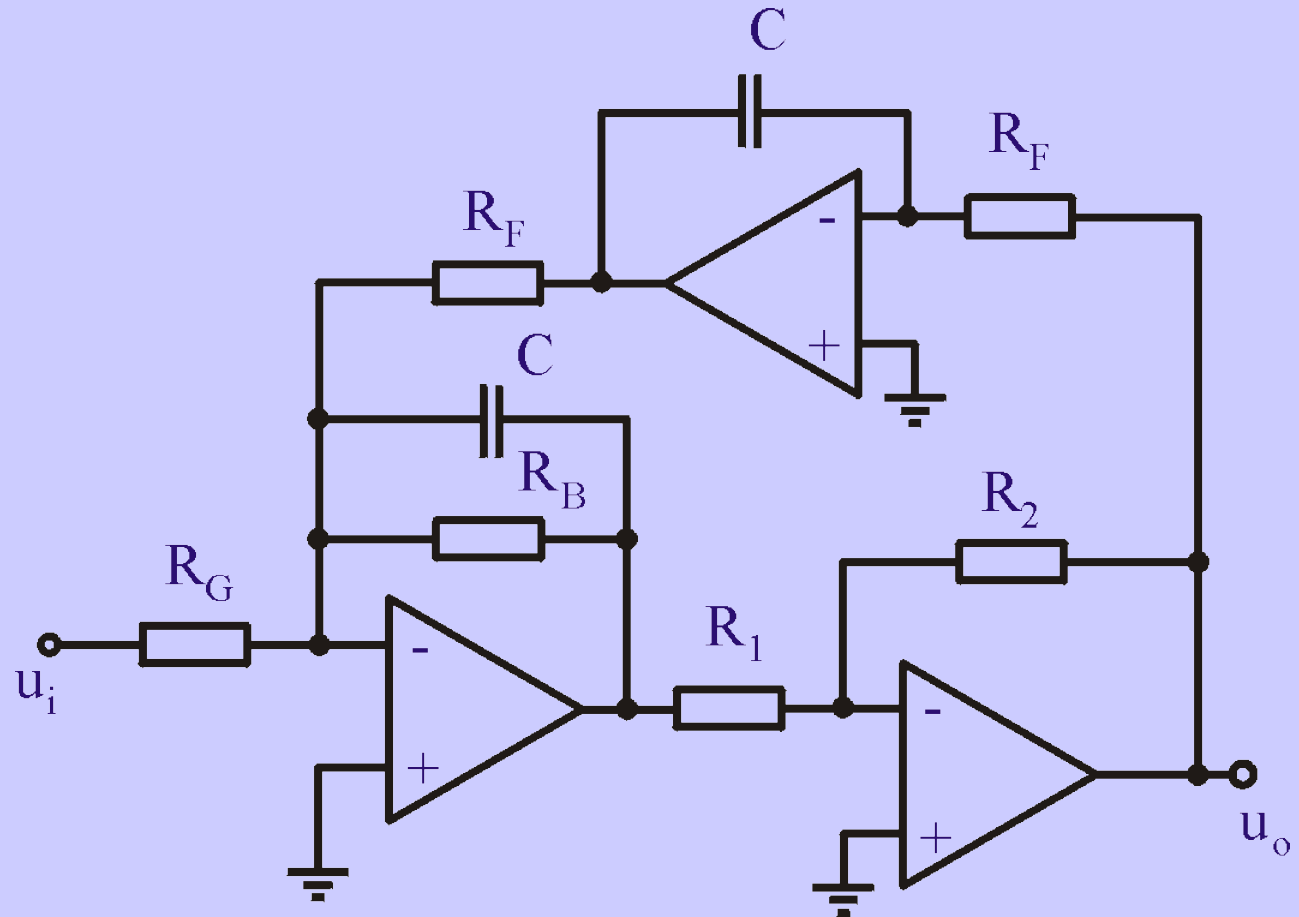
Biquad kaistanpäästösuodatin

- vastuksilla R_F voidaan säätää keskitaajuutta muuttamatta taajuuskaistan leveyttä

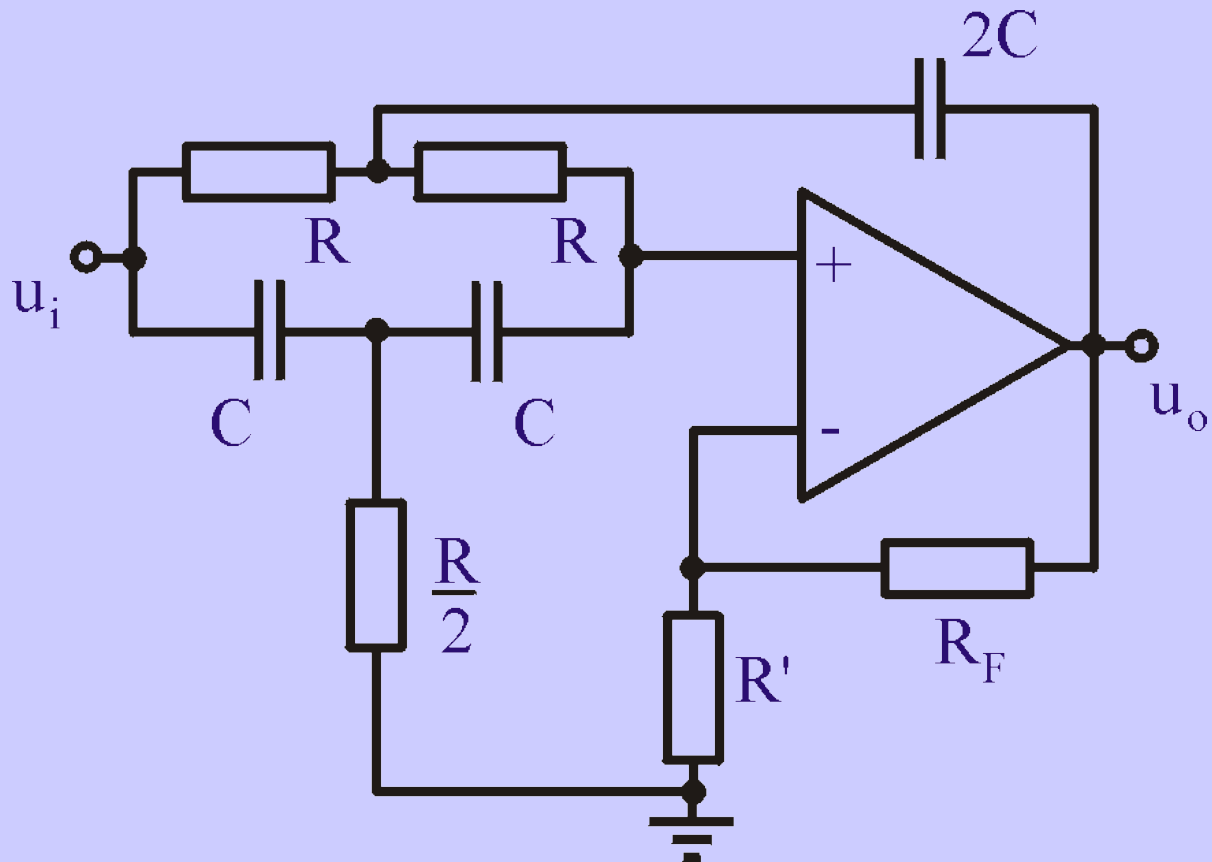
$$f_0 = \frac{1}{2\pi R_F C} ,$$

$$\Delta f = \frac{1}{2\pi R_B C}$$

$$A_V = \frac{R_B}{R_G} .$$

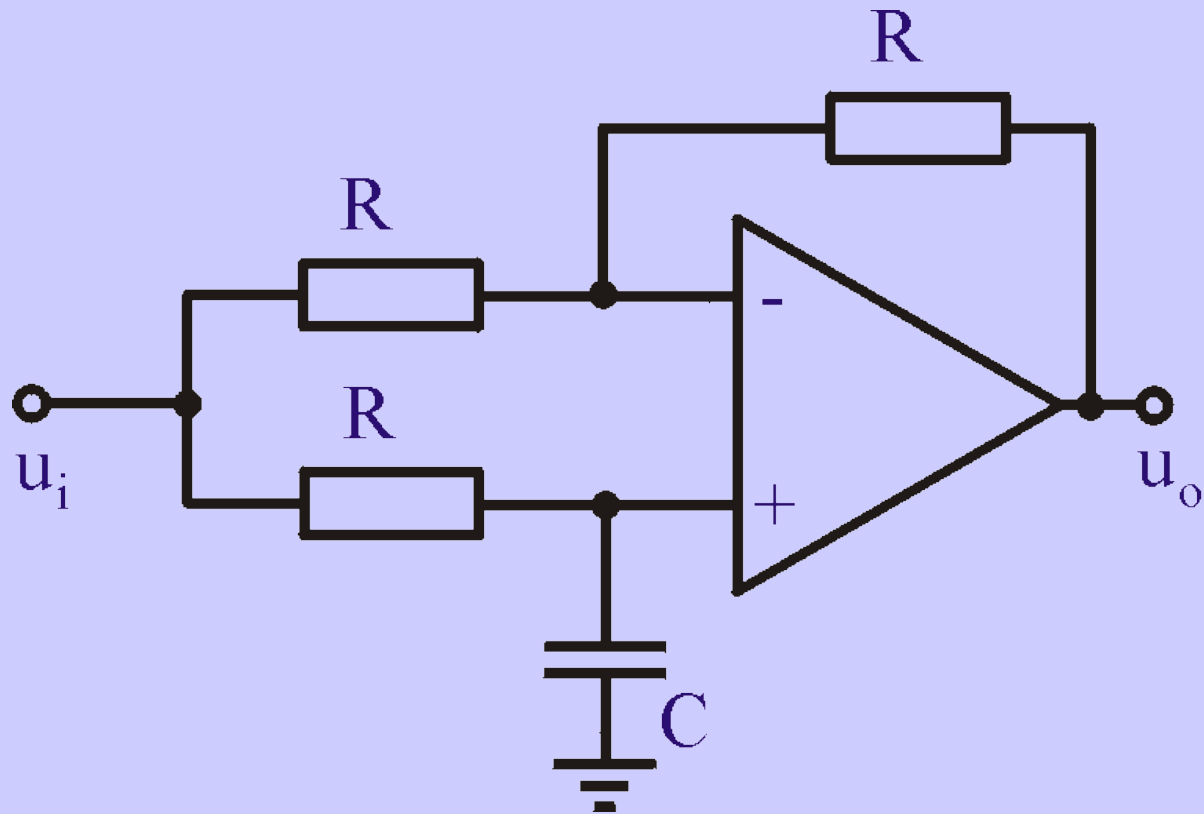


Kaistanestosuodatin, kaksois-T -kytkentä



$$f_0 = \frac{1}{2\pi RC}$$

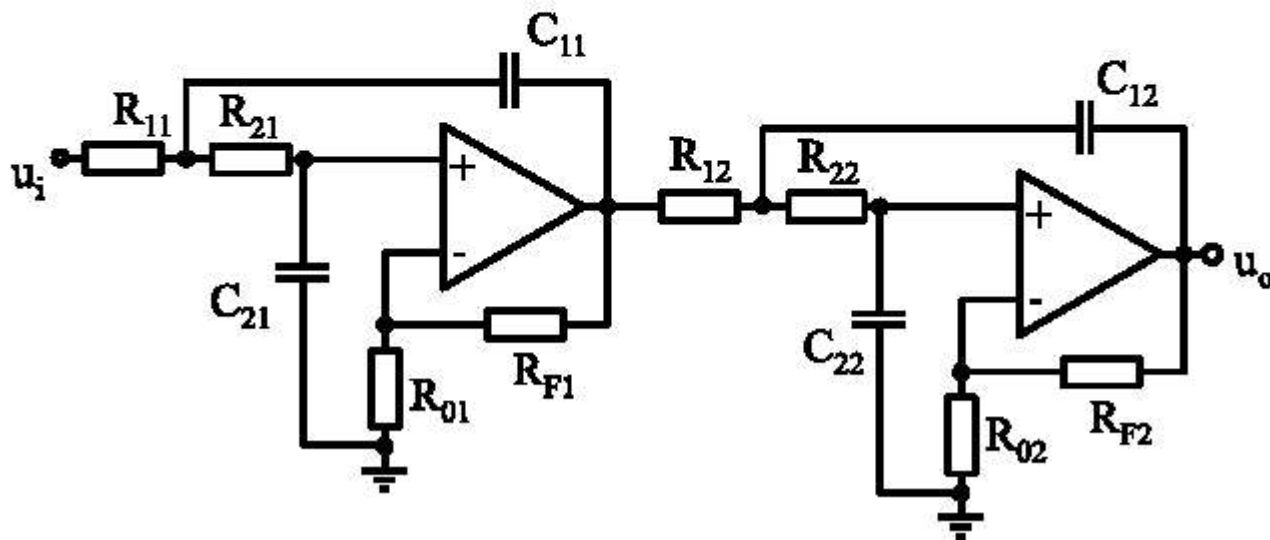
Vaiheen muuttaminen, *all-pass* -suodatin

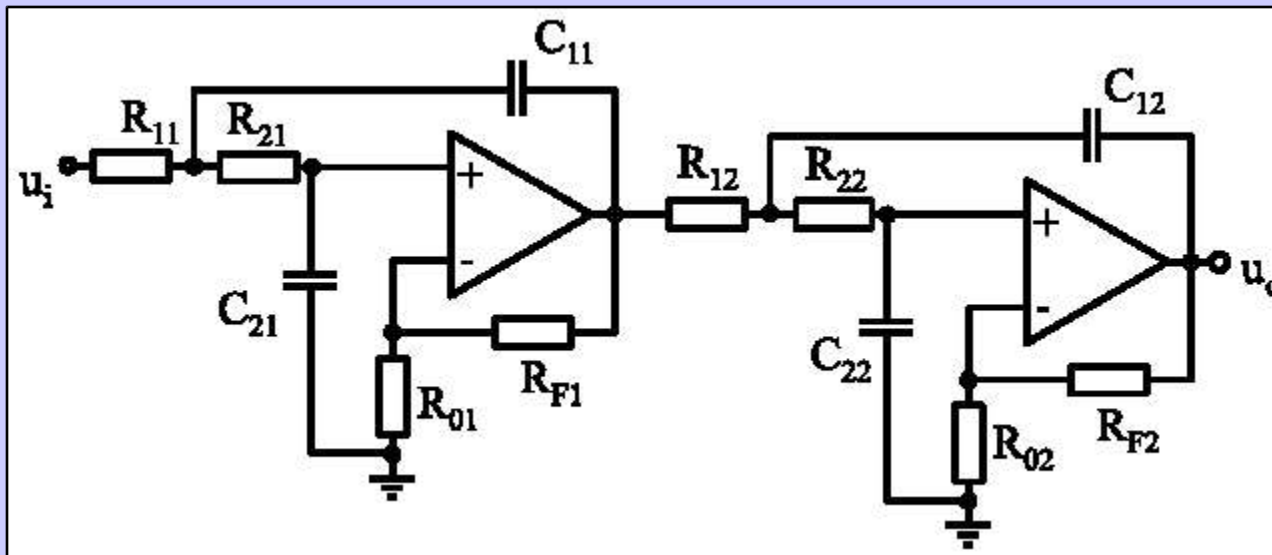


Esimerkki 5.2

Mitoita 4. kertaluvun Butterworthin ja Besselin alipäästösuodattimet (yläraajataajuus 20 kHz). Valitse kapasitanssiarvot taulukon 1.3 sarakkeesta E12 ja vastusarvot sarakkeesta E96 (1% vastukset).

Tarvitaan kaksi 2. kertaluvun suodatinta peräkkäin:





Butterworth

1. aste, valitaan esim. $C_{11} = C_{21} = 820 \text{ pF}$, jolloin

$$R_{11} = R_{21} = \frac{1}{2\pi \cdot 20\text{kHz} \cdot 820\text{pF}} = 9705\Omega \approx 9,76\text{k}\Omega \quad (1\% \text{ vastus})$$

ja vahvistus ($A_{V1} = 1,152$ taulukosta 5.1)

olkoon $R_{01} = 10\text{k}\Omega$, jolloin $R_{F1} = (A_{V1} - 1) \cdot R_{01} = 1,52\text{k}\Omega \approx 1,54\text{k}\Omega$.

2. aste, vastus- ja kapasitanssiarvot voidaan asettaa samoiksi eli

$$C_{12} = C_{22} = 820 \text{ pF} \text{ ja } R_{12} = R_{22} = 9,76 \text{ k}\Omega$$

ja vahvistus ($A_{V2} = 2,235$)

olkoon $R_{02} = 10\text{k}\Omega$, jolloin $R_{F2} = (A_{V2} - 1) \cdot R_{02} = 12350\Omega \approx 12,4\text{k}\Omega$.

Bessel

1. aste, valitaan esim. $C_{11} = C_{21} = 820 \text{ pF}$, jolloin (kaavasta 5.35)

$$R_{11} = R_{21} = \frac{1}{2\pi \cdot 1,432 \cdot 20\text{kHz} \cdot 820\text{pF}} = 6777\Omega \approx 6,81\text{k}\Omega$$

ja vahvistus ($A_{V1} = 1,084$)

olkoon $R_{01} = 10\text{k}\Omega$, jolloin $R_{F1} = (A_{V1} - 1) \cdot R_{01} = 840\Omega \approx 845\Omega$.

2. aste, valitaan esim. $C_{12} = C_{22} = 820 \text{ pF}$, jolloin

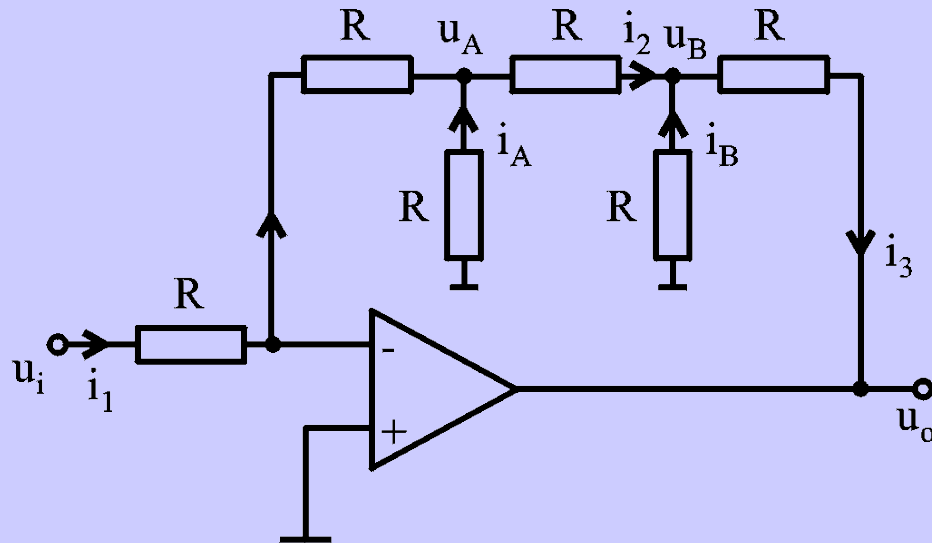
$$R_{12} = R_{22} = \frac{1}{2\pi \cdot 1,606 \cdot 20\text{kHz} \cdot 820\text{pF}} = 6043\Omega \approx 6,04\text{k}\Omega$$

ja vahvistus ($A_{V2} = 1,759$)

olkoon $R_{02} = 10\text{k}\Omega$, jolloin $R_{F2} = (A_{V2} - 1) \cdot R_{02} = 7590\Omega \approx 7,5\text{k}\Omega$.



Exempel



$$i_1 = \frac{u_i}{R}, u_A = -Ri_1 = -u_i$$

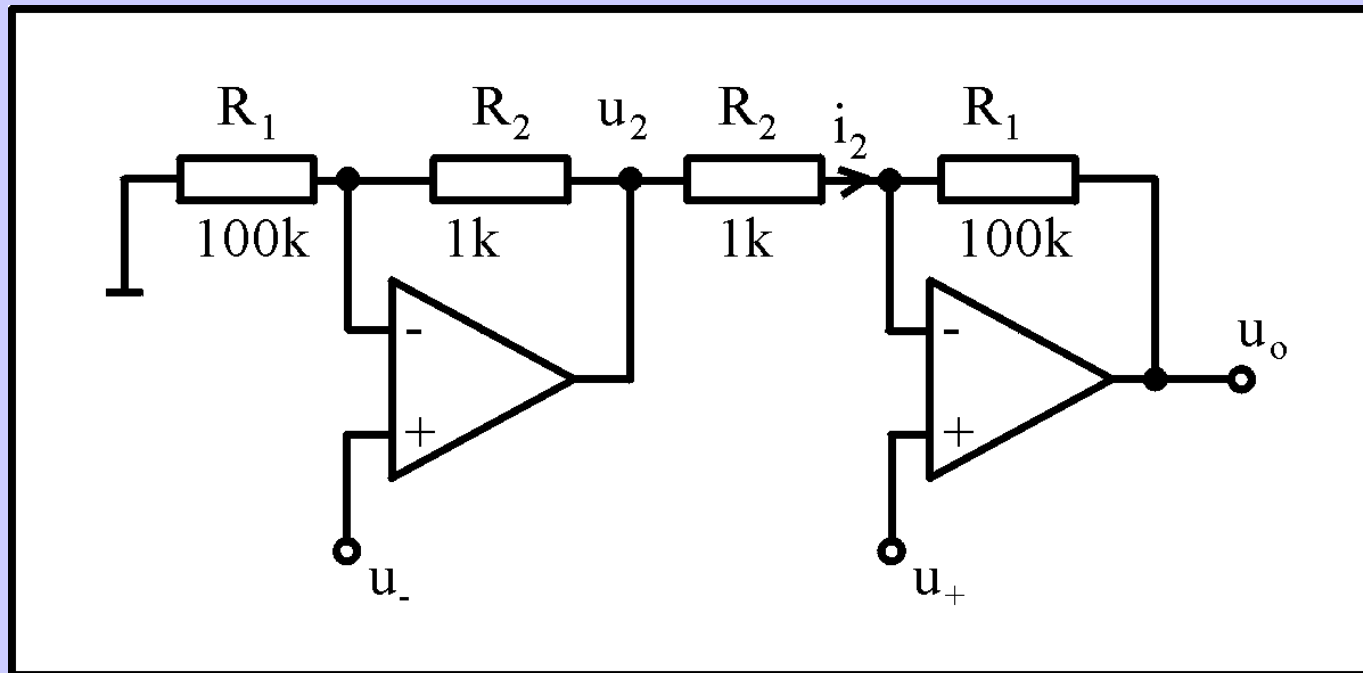
$$i_A = \frac{u_i}{R}, i_2 = i_1 + i_A = 2i_1 = \frac{2u_i}{R}$$

$$u_B = u_A - R(i_1 + i_A) = u_A - 2u_i = -3u_i$$

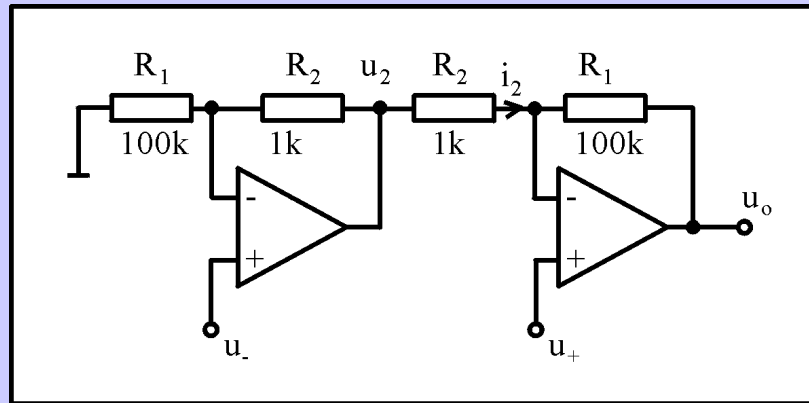
$$i_B = \frac{u_B}{R} = -\frac{3u_i}{R}, u_o = u_B - R(i_2 + i_B) = -3u_i - 2u_i - 3u_i = -8u_i$$

$$A_V = \frac{u_o}{u_i} = -8$$

Exempel



Exempel



$$u_1 = \left(1 + \frac{R_2}{R_1}\right) u_-$$

$$u_o = u_+ - i_2 R_1 =$$

$$u_+ - \frac{u_1 - u_+}{R_2} R_1 = u_+ - \left(1 + \frac{R_2}{R_1}\right) \frac{u_- - u_+}{R_2} R_1 =$$

$$u_+ - \left(\frac{R_1}{R_2} + 1\right) u_- + \frac{R_1}{R_2} u_+ = (u_+ - u_-) \cdot \left(1 + \frac{R_1}{R_2}\right)$$

$$A_v = \frac{100k}{1k} + 1 = 101$$

